

Quantum Mechanics from Admissible Probabilistic Composition

A Reconstruction via Unistochastic Matrices

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Preface

Scope. This book does not conclude that complex numbers are metaphysically necessary, uniquely forced, or the only mathematics capable of describing quantum phenomena. Every result ruling out $\mathbb{R}$ does so conditional on stated assumptions — a specific admissibility constraint (Chapters 5–6), or an explicitly adopted principle such as local tomography (Chapter 8, itself flagged throughout as assumed, not derived). Chapter 18 collects every such assumption in one place, exactly four of them, and Chapter 18’s §18.4 and Chapter 19 discuss published 2025–2026 work showing that changing the assumption about how systems compose reproduces every quantum prediction using only real numbers. A reader who takes away “this book proves the universe requires complex numbers” has taken away a claim this book explicitly argues against.

This book asks how much of quantum mechanics can be recovered from constraints on admissible probabilistic transformations, rather than introduced by postulate. The starting point is elementary: a transition matrix built from squared amplitudes, $B_{ij} = |U_{ij}|^2$, is the unistochastic object at the center of Życzkowski, Kuś, Słomczyński, and Sommers’s [13] work and of Barandes’s stochastic–quantum correspondence [5]. The question this book pursues is what happens if that object is treated not as a curiosity about quantum probability, but as the thing to be explained: under what admissibility constraints on probabilistic data does a coherent unitary lift become necessary, and how much of the standard formalism follows once it does?

Two methodological commitments run through every chapter.

Reconstruction versus recovery. A reconstruction claim is proved from constraints already established earlier in this book. A recovery claim shows that a reconstructed structure reproduces standard quantum mechanics, checked directly against Sakurai–Napolitano’s [1] *Modern Quantum Mechanics* and Gross’s [3] *Advanced Quantum Mechanics* lecture notes. (The older Sakurai *Advanced Quantum Mechanics* [2] was part of the originally intended reference corpus but survives only as a scanned, non-text-searchable copy; it is listed for completeness but was not directly quotable, and no claim in this book depends on it.) The two are never allowed to blur into each other: a result that merely matches the textbook formalism is not thereby derived, and a result derived from this book’s admissibility logic is not thereby the same claim as its textbook counterpart until the match is checked explicitly.

The assumption ledger. Each chapter closes with a boxed ledger stating plainly what was established and what was not — including, where relevant, exactly which new physical or operational assumption a result costs. This is not a stylistic tic. Fifty-some pages into a successful reconstruction it becomes easy to feel that quantum mechanics has been derived outright; Chapter 13 and Chapter 18 exist specifically to check that feeling against the record, and the per-chapter ledgers are what make that final audit possible to write honestly rather than from memory.

The book has three parts. Part I asks why bare probability data is insufficient on its own

— first sequentially (composing transformations loses information under projection) and then contextually (three individually admissible measurement contexts can fail to admit any single real realization at once, the sharpest form of which is a minimal graph-theoretic witness, the triangle). Part II asks what composing separate systems, rather than composing or contextualizing a single one, forces and distinguishes among candidate scalar fields — real, complex, quaternionic — culminating in a single matrix whose entanglement status depends on which field is doing the composing. Part III returns to the standard formalism and asks how much of it can be recovered from what Parts I and II established, closing with an explicit accounting of every assumption the recovery required.

Four such assumptions appear in the whole book, no more and no fewer, and they are collected in one place in Chapter 18. The reader who wants to know exactly what standard quantum mechanics cost, on this route, need only turn there.

One methodological habit recurs often enough to name here rather than leave implicit: whenever an object in this book appears indispensable — a particular matrix representation, a particular composition rule, a particular coordinatization — the question asked is always whether it is a genuine admissibility constraint or merely one convenient way of writing a deeper invariant structure. Complex numbers themselves are not exempt from this question; nor is the specific Kronecker-product convention used to combine systems throughout Parts I and II (§18.4 returns to this point directly, in light of recent literature showing that composition, not the scalar field alone, carries much of the weight usually attributed to i itself). A representation earns its place by what it lets the reader compute or check — not by familiarity, and not by having no visible alternative.

Part I

Single-System Foundations

Chapter 1

The Probability Simplex

1.1 Distinguishable Alternatives

Let $X = \{1, \dots, n\}$ be a finite set of operationally distinguishable alternatives — outcomes of some fixed measurement procedure, prior to any further structure. A state of knowledge about which alternative obtains is a probability distribution

$$p = (p_1, \dots, p_n), \quad p_i \geq 0, \quad \sum_i p_i = 1, \quad (1.1)$$

i.e. a point of the standard probability simplex $\Delta_{n-1} \subset \mathbb{R}^n$. Δ_{n-1} is a convex set with n extreme points, the vertices e_i (certainty that alternative i obtains); every $p \in \Delta_{n-1}$ is a convex combination of vertices.

Remark 1.1 (Constraint before content). Nothing has been said yet about what alternative i is — a detector click, a coin face, a measurement outcome. The simplex is defined purely by the admissibility constraints (nonnegativity, normalization) on the space of possible states of knowledge. This is the methodological stance the whole book follows: characterize the admissible transformations on a space before assigning any ontology to the space's points. The simplex is therefore a common starting point for many probabilistic theories, classical and quantum alike; what distinguishes quantum mechanics from every other such theory has not yet entered anywhere in this chapter. The same stance applies to notation, not only to ontology: a later instance of this principle (§18.4) asks whether the complex numbers used from Chapter 3 onward are themselves an admissibility constraint or one convenient coordinatization of a deeper real structure — the question is posed here in its first, simplest form.

1.2 Stochastic Maps

A transformation between two such state spaces (or a single space to itself, $X \rightarrow X$) is a stochastic map: a matrix M with

$$M_{ij} \geq 0, \quad \sum_i M_{ij} = 1 \quad \text{for every } j, \quad (1.2)$$

acting on distributions by $p'_i = \sum_j M_{ij} p_j$. Column-normalization ensures $p' \in \Delta_{n-1}$ whenever $p \in \Delta_{n-1}$.

Definition 1.2 (Deterministic maps). A stochastic map M is deterministic if every column is a standard basis vector e_i , i.e. M implements a function $f : X \rightarrow X$ with $M_{ij} = \delta_{i,f(j)}$.

Proposition 1.3. The deterministic stochastic maps are exactly the extreme points of the convex set of stochastic maps on X .

Proof. Every deterministic M is extreme: if $M = \frac{1}{2}(M_1 + M_2)$ for stochastic M_1, M_2 , then in the column where M has a 1-entry, both M_1, M_2 must also equal 1 there (as in the Birkhoff–von Neumann argument of Theorem 2.2, using only column normalization and nonnegativity — row normalization is not needed here), forcing $M_1 = M_2 = M$ in that column, and since this holds for every column, $M_1 = M_2 = M$.

Conversely, if M is not deterministic, some column j has at least two nonzero entries $M_{i_1j}, M_{i_2j} \in (0, 1)$. Let $\varepsilon > 0$ be less than $\min(M_{i_1j}, M_{i_2j}, 1 - M_{i_1j}, 1 - M_{i_2j})$ and define $M_{\pm}$ by adding $\pm\varepsilon$ to column j 's i_1 -entry and $\mp\varepsilon$ to its i_2 -entry, leaving every other entry of M unchanged. Column j still sums to 1 (the two perturbations cancel) and all entries remain in $[0, 1]$, so $M_{\pm}$ are stochastic; $M = \frac{1}{2}(M_+ + M_-)$ with $M_+ \neq M_-$, so M is not extreme. $\square$

1.3 Composition, Coarse-Graining, and Information Loss

Stochastic maps compose by matrix multiplication: applying M_1 then M_2 gives M_2M_1 , again stochastic. Two features of this composition matter for everything that follows.

Coarse-graining. A deterministic map that merges several alternatives of a larger space into fewer alternatives of a smaller one is a coarse-graining; it is generally not invertible; information about which of the merged alternatives held is lost.

Convex mixtures. If M_1, M_2 are stochastic maps on the same pair of spaces, so is $\lambda M_1 + (1 - \lambda)M_2$ for $\lambda \in [0, 1]$: the stochastic maps between two fixed spaces themselves form a convex set.

Remark 1.4. Nothing in this chapter has yet distinguished admissible stochastic maps from arbitrary ones beyond the bare normalization constraint. Chapter 2 imposes the further constraint of reversibility (bistochasticity), which is where the geometry relevant to quantum mechanics begins to appear.

Ledger 1.5 (End of Chapter 1). Established: the probability simplex and stochastic maps as the starting admissibility structure, prior to any notion of amplitude, phase, or reversibility. Not yet established: any distinction between classical and quantum transformations — that distinction requires the further restriction to bistochastic (Chapter 2) and then unistochastic (Chapter 3) maps.

Chapter 2

Bistochastic Matrices and the Birkhoff Polytope

2.1 Bistochasticity

A stochastic map M (Chapter 1) is bistochastic (doubly stochastic) if, in addition to column sums, its row sums also equal one:

$$B_{ij} \geq 0, \quad \sum_i B_{ij} = 1, \quad \sum_j B_{ij} = 1. \quad (2.1)$$

Write $\text{DS}(n)$ for the set of $n \times n$ bistochastic matrices. Row normalization is the additional admissibility constraint motivating this chapter: a bistochastic map sends the uniform distribution to itself, and (as Theorem 2.2 below shows) is built entirely from maps that merely relabel alternatives.

2.2 The Birkhoff Polytope

Bistochastic matrices have long been treated as the natural probabilistic analogue of reversible transformations: permutations are the only transformations on a finite set that never lose distinguishability, and a bistochastic matrix is, by the theorem below, always assembled from a mixture of them. That association motivates why this particular restriction of $\text{DS}(n)$ — not some other subset — is the one worth naming and studying before narrowing the admissible class further.

Definition 2.1. A permutation matrix P_σ , for $\sigma \in S_n$, has $(P_\sigma)_{ij} = \delta_{i,\sigma(j)}$. Every permutation matrix is bistochastic.

Theorem 2.2 (Birkhoff–von Neumann). $\text{DS}(n)$ is a convex polytope (the Birkhoff polytope) whose extreme points are exactly the $n!$ permutation matrices. Consequently every $B \in \text{DS}(n)$ can be written as a convex combination

$$B = \sum_{\sigma \in S_n} c_\sigma P_\sigma, \quad c_\sigma \geq 0, \quad \sum_\sigma c_\sigma = 1, \quad (2.2)$$

though not, in general, uniquely.

Proof. $\text{DS}(n)$ is defined by finitely many linear equalities (row/column sums) and inequalities ($B_{ij} \geq 0$) in a bounded region of $\mathbb{R}^{n^2}$, hence is a convex polytope.

Permutation matrices are extreme. Suppose $P_\sigma = \frac{1}{2}(M_1 + M_2)$ for $M_1, M_2 \in \text{DS}(n)$. Wherever $(P_\sigma)_{ij} = 0$, nonnegativity of M_1, M_2 forces $(M_1)_{ij} = (M_2)_{ij} = 0$. Wherever $(P_\sigma)_{ij} = 1$,

each row of M_1 (resp. M_2) sums to 1 with all entries ≥ 0 , so $(M_1)_{ij} \leq 1$ and likewise for M_2 ; averaging to 1 with both terms ≤ 1 forces $(M_1)_{ij} = (M_2)_{ij} = 1$. Hence $M_1 = M_2 = P_\sigma$.

Every extreme point is a permutation matrix. Let $B \in \text{DS}(n)$ be arbitrary and not a permutation matrix, and let G_B be the bipartite support graph on row-vertices and column-vertices with an edge (i, j) whenever $B_{ij} > 0$. Every vertex of G_B has degree ≥ 1 (each row/column sums to 1 > 0), and since B is not a permutation matrix, some vertex has degree ≥ 2 . A finite graph with a vertex of degree ≥ 2 and no vertex of degree 0 contains a cycle; since G_B is bipartite, this cycle has even length and alternates rows and columns,

$$i_1, j_1, i_2, j_2, \dots, i_k, j_k, i_1, \quad B_{i_1 j_1} > 0, \quad B_{i_{k+1} j_1} > 0. \quad (2.3)$$

Let $\varepsilon > 0$ be smaller than every entry of B along the cycle, and define Δ by alternating $+\varepsilon, -\varepsilon$ along the cycle edges (zero elsewhere). Because the pattern alternates sign around a closed cycle, every row and column sum of Δ is zero, so $B \pm \Delta$ are again nonnegative (by choice of ε) with the same row and column sums as B : both lie in $\text{DS}(n)$. Since $B = \frac{1}{2}[(B + \Delta) + (B - \Delta)]$ and $\Delta \neq 0$, B is not extreme. Contrapositively, every extreme point is a permutation matrix.

The final claim (every $B \in \text{DS}(n)$ is a convex combination of extreme points) is the Krein–Milman property of a finite-dimensional compact convex set applied to the polytope $\text{DS}(n)$. $\square$

Remark 2.3. Theorem 2.2 already illustrates the book’s recurring pattern: a physically motivated constraint (bistochasticity) carves out a polytope whose extreme structure (permutations) is exactly the reversible, information-preserving special case, with everything else in the polytope representable as a mixture of reversible maps. This is the classical precursor to the much sharper restriction imposed by orthostochasticity and unistochasticity in Chapter 3, where the analogous extreme structure is no longer merely a finite set of permutations but a continuum requiring genuine geometric (and, past $n = 2$, non-convex) analysis.

2.3 Majorization

Bistochastic maps have a further characterization in terms of mixing: applying one can only make a distribution more uniform, never less.

Definition 2.4 (Majorization). For $p, q \in \Delta_{n-1}$, write $q \prec p$ (“ q is majorized by p ”) if, after sorting both in decreasing order,

$$\sum_{i=1}^k q_i^\downarrow \leq \sum_{i=1}^k p_i^\downarrow \quad \text{for every } k = 1, \dots, n-1 \quad (2.4)$$

(with equality automatic at $k = n$).

Theorem 2.5 (Hardy–Littlewood–Pólya). $q \prec p$ if and only if $q = Bp$ for some $B \in \text{DS}(n)$.

Proof. ($\Leftarrow$) Suppose $q = Bp$ for $B \in \text{DS}(n)$. Fix $k \in \{1, \dots, n\}$ and let S be any set of k indices. Then

$$\sum_{i \in S} q_i = \sum_{i \in S} \sum_j B_{ij} p_j = \sum_j w_j p_j, \quad w_j := \sum_{i \in S} B_{ij} \in [0, 1], \quad \sum_j w_j = k, \quad (2.5)$$

the last equality because each column of B sums to 1 and $|S| = k$. Among all weight vectors w with entries in $[0, 1]$ summing to k , $\sum_j w_j p_j$ is maximized by placing weight 1 on the k largest entries of p (an elementary rearrangement argument), giving

$$\sum_{i \in S} q_i \leq \sum_{i=1}^k p_i^\downarrow \quad (2.6)$$

for every S of size k ; in particular for S the indices of the k largest entries of q , this is exactly $\sum_{i=1}^k q_i^\downarrow \leq \sum_{i=1}^k p_i^\downarrow$. Since k was arbitrary, $q \prec p$.

($\Rightarrow$) Suppose $q \prec p$. A T -transform on indices (i, j) with parameter $t \in [0, 1]$ is the bistochastic matrix $T = I - t(e_i - e_j)(e_i - e_j)^\top$, a convex combination $t P_{(ij)} + (1 - t) I$ of the identity and the transposition (ij) ; applying it moves mass between coordinates i, j while preserving their sum, replacing (p_i, p_j) by a pair strictly between them. The classical construction (Muirhead) produces q from p by a finite sequence of such transfers, each moving mass from a coordinate where p (or its current partially-transformed image) exceeds the corresponding partial sums of q to one where it falls short, terminating once all partial-sum inequalities defining $q \prec p$ are saturated. Each T -transform lies in $\text{DS}(n)$, and $\text{DS}(n)$ is closed under matrix multiplication (row/column sums of 1 and nonnegativity are both preserved by composing two such maps), so the composite $B = T_m \cdots T_1$ of the finitely many transforms used again lies in $\text{DS}(n)$, and $q = Bp$ by construction. $\square$

Corollary 2.6 (Entropy is not decreased by mixing). If $q = Bp$ for $B \in \text{DS}(n)$, then $H(q) \geq H(p)$, where $H(p) = -\sum_i p_i \log p_i$ is the Shannon entropy.

Remark 2.7. Corollary 2.6 follows from Theorem 2.5 together with the Schur-concavity of H : entropy increase under bistochastic evolution is a structural consequence of majorization geometry, not a thermodynamic postulate added on top of the probabilistic framework. This is the classical, pre-quantum ancestor of the entropy behavior discussed later under coarse-graining and decoherence.

Ledger 2.8 (End of Chapter 2). Established: bistochastic matrices form a polytope generated by permutations (Theorem 2.2); bistochastic evolution is exactly majorization-compatible evolution (Theorem 2.5), and cannot decrease Shannon entropy (Corollary 2.6).

Not yet established: any restriction singling out orthostochastic orunistochastic matrices within $\text{DS}(n)$ — that restriction, and its non-convex geometry for $n \geq 3$, is the subject of Chapter 3.

Chapter 3

Orthostochastic and Unistochastic Admissibility

3.1 Two Notions of Admissibility

A bistochastic matrix $B \in \text{DS}(n)$ is orthostochastic if $B_{ij} = O_{ij}^2$ for some real orthogonal O , and unistochastic if $B_{ij} = |U_{ij}|^2$ for some unitary U . Write $\text{OS}(n)$ and $\text{US}(n)$ for the respective sets. Since a real orthogonal matrix is a special case of a unitary one, $\text{OS}(n) \subseteq \text{US}(n) \subseteq \text{DS}(n)$ always.

3.2 The $n = 2$ Degeneracy Theorem

Theorem 3.1 ($n = 2$ coincidence). $\text{OS}(2) = \text{US}(2) = \text{DS}(2)$.

Proof. Every $B \in \text{DS}(2)$ has the form $B(t) = \begin{pmatrix} t & 1-t \\ 1-t & t \end{pmatrix}$, $t \in [0, 1]$, by the row/column sum constraints. Setting $t = \cos^2 \theta$, both

$$O(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \quad \text{and} \quad U(\theta) = O(\theta) \quad (3.1)$$

realize $B(t)$: $O_{ij}(\theta)^2 = |U_{ij}(\theta)|^2 = B(t)_{ij}$. Since a real orthogonal matrix is already unitary, this single realization witnesses membership in both $\text{OS}(2)$ and $\text{US}(2)$ simultaneously. $\square$

Remark 3.2. This is a statement about a single transition matrix: at $n = 2$, B alone cannot distinguish a real from a complex (or, degenerately, quaternionic) lift. It does not imply that two-outcome systems are effectively classical — sequential composition of lifts still exposes phase information not visible in any single B (Chapter 4), and joint consistency across multiple contexts still obstructs a purely real realization (Chapter 6). Membership of one B and coherent composition of its lifts are distinct questions.

3.3 The $n \geq 3$ Incompleteness Theorem

At $n = 2$ every unistochastic matrix is orthostochastic. This fails already at $n = 3$, and by a strikingly concrete counterexample.

Theorem 3.3 (Unistochastic strictly larger than orthostochastic, $n = 3$). $\text{OS}(3) \subsetneq \text{US}(3)$.

Proof. Let U be the 3×3 discrete Fourier transform matrix,

$$U = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix}, \quad \omega = e^{2\pi i/3}. \quad (3.2)$$

U is unitary, and $\Gamma_{ij} = |U_{ij}|^2 = 1/3$ for every i, j , so $\Gamma \in \text{US}(3)$.

Suppose $\Gamma \in \text{OS}(3)$ as well, i.e. $\Gamma_{ij} = O_{ij}^2$ for some real orthogonal O . Then every entry of O satisfies $O_{ij}^2 = 1/3$, so $O_{ij} = \pm 1/\sqrt{3}$ for all nine entries. Orthogonality of O requires $OO^T = I$; in particular every pair of distinct rows must be orthogonal, i.e. $\sum_k O_{ik}O_{jk} = 0$ for $i \neq j$. Writing $O = \frac{1}{\sqrt{3}}H$ with H a ± 1 sign matrix, this is exactly the condition that H be a 3×3 Hadamard matrix ($HH^T = 3I$). No such matrix exists: an exhaustive check of all $2^9 = 512$ sign patterns finds zero satisfying $HH^T = 3I$, consistent with the classical fact that Hadamard matrices exist only for orders 1, 2, and multiples of 4. Contradiction; hence $\Gamma \notin \text{OS}(3)$. $\square$

Corollary 3.4. Complex amplitudes are structurally necessary to realize every unistochastic matrix once $n \geq 3$: $\text{OS}(n) \subsetneq \text{US}(n)$ for all $n \geq 3$.

Scope. This is a fact about one mathematical object — a specific unistochastic matrix, and the matrices that can or cannot reproduce it — proved without any assumption beyond ordinary linear algebra. It is not yet a claim about physical reality’s scalar field: nothing here has connected “this matrix” to “a physical measurement,” and that connection is made only later, with its own explicit costs (Chapter 14’s Assumption 14.2). Nor does it settle which representation of the underlying real structure is required — Chapter 19 shows the algebraic content normally written with i can equally be written with real 2×2 matrices; what this corollary shows is that the standard orthostochastic (real, entrywise-squared) representation specifically cannot reproduce this matrix, not that no real structure whatsoever could, once the representation itself is allowed to change.

Remark 3.5. Theorem 3.3 is logically independent of the multi-context cycle obstruction of Chapter 6: it is a statement about a single bistochastic matrix failing to admit any real lift at all, whereas the cycle obstruction concerns three individually-liftable edges failing to admit a jointly consistent real lift. Both point toward the same conclusion via different mechanisms, and the book should keep them distinct rather than treat one as a restatement of the other.

Ledger 3.6 (End of Chapter 3). Established: at $n = 2$, orthostochastic and unistochastic coincide, so no single transition matrix distinguishes real from complex lifts (Theorem 3.1); at $n = 3$, an explicit unistochastic matrix (the DFT matrix’s modulus-square) admits no real lift whatsoever (Theorem 3.3), via the nonexistence of a 3×3 Hadamard matrix.

Not yet established: whether every unistochastic matrix that fails to be orthostochastic fails for the same reason (a general structural criterion, rather than a single worked example); the relationship, if any, between this single-matrix obstruction and the cycle obstruction of Chapter 6.

Chapter 4

The Composition Obstruction

Chapter 3 showed that some matrices lack a real lift at all. This chapter asks a different, logically prior question: even when a lift does exist, does composing two lifted transformations and then projecting to probabilities give the same answer as composing the probabilities directly? The obstruction that follows does not depend on anything from Chapter 3 failing — it appears even when every matrix involved is perfectly well-behaved, which is exactly what makes it a different kind of problem from a missing lift.

4.1 Amplitude-Level versus Probability-Level Composition

Let U_1, U_2 be unitary matrices and define the modulus-square map

$$\pi(U)_{ij} = |U_{ij}|^2. \quad (4.1)$$

Composing at the amplitude level and then projecting gives

$$|(U_2 U_1)_{ij}|^2 = \left| \sum_k (U_2)_{ik} (U_1)_{kj} \right|^2 = \sum_k |(U_2)_{ik}|^2 |(U_1)_{kj}|^2 + \sum_{k \neq \ell} (U_2)_{ik} (U_1)_{kj} \overline{(U_2)_{i\ell} (U_1)_{\ell j}}. \quad (4.2)$$

The first term on the right is exactly the classical (probability-level) stochastic composition $(B_2 B_1)_{ij}$, where $B_k = \pi(U_k)$. The second term is discarded entirely by probability-level composition. The obstruction is not the existence of interference itself, but the failure of probability projection to preserve the composition law.

Proposition 4.1 (Non-homomorphism of π). In general, $\pi(U_2 U_1) \neq \pi(U_2) \pi(U_1)$.

Example 4.2 (A 2×2 counterexample). Let $U_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ (the real Hadamard matrix) and $U_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}$. Direct computation gives $U_2 U_1 = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$, so $\pi(U_2 U_1) = I$ — the identity permutation matrix. But $\pi(U_1) = \pi(U_2) = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ (every entry of both U_1, U_2 has modulus $\frac{1}{\sqrt{2}}$), and this matrix is idempotent, so $\pi(U_2) \pi(U_1) = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \neq I$. Composing at the probability level turns a perfectly deterministic outcome (identity) into maximal uncertainty (uniformly mixed).

Example 4.3 (A 3×3 counterexample, reusing Chapter 3's DFT matrix). Let F be the 3×3 discrete Fourier transform matrix of Theorem 3.3, for which $\pi(F)$ is the uniform matrix with every entry $\frac{1}{3}$. Direct computation gives $F^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$, a genuine permutation matrix, so $\pi(F^2) = F^2$ itself: another deterministic outcome. But $\pi(F)$ is idempotent (every entry of $\pi(F) \pi(F)$ is a sum of three terms $\frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9}$, giving $\frac{1}{3}$ again), so $\pi(F)^2 = \pi(F) \neq \pi(F^2)$: again, a deterministic amplitude-level outcome is replaced by maximal uncertainty once probabilities are composed instead of amplitudes.

Remark 4.4. Equation (4.2) licenses the informal identification

$$\text{interference} \sim \text{information required for coherent composition but erased by } \pi.$$

This should be read as an interpretation of the cross terms, not as a definition; the precise claim is Proposition 4.1.

4.2 The Real Counterexample

The obstruction identified above does not select complex amplitudes over real ones. Let O_1, O_2 be real orthogonal matrices and define the orthostochastic projection $\pi_{\mathbb{R}}(O)_{ij} = O_{ij}^2$. The identical expansion

$$(O_2 O_1)_{ij}^2 = \sum_k (O_2)_{ik}^2 (O_1)_{kj}^2 + 2 \sum_{k < \ell} (O_2)_{ik} (O_1)_{kj} (O_2)_{i\ell} (O_1)_{\ell j} \quad (4.3)$$

produces the same cross-term structure, and $\pi_{\mathbb{R}}(O_2 O_1) \neq \pi_{\mathbb{R}}(O_2) \pi_{\mathbb{R}}(O_1)$ in general.

Ledger 4.5 (End of Chapter 4). Established: probability-level composition discards information present in amplitude-level composition; the discarded information generates interference cross terms; a coherent lift (real or complex) restores compositional closure.

Not established: that the lift must be complex. See Proposition 4.1 and §4.2:

interference $\not\Rightarrow \mathbb{C}$.

This entry remains open until Chapter 6.

Chapter 5

Context Graphs and Cycle Consistency

Chapter 4 asked whether projecting before composing preserves the composition law for a fixed pair of lifts. This chapter asks a different question entirely: given several independently observed transitions between measurement contexts, can they all have come from one consistent global assignment of frames? The first question is about a single, chosen composition; the second is about whether many separately admissible observations are mutually compatible at all. Nothing about Chapter 4’s obstruction is assumed or needed here.

5.1 Measurement Contexts as Vertices, Transitions as Edges

Let $G = (V, E)$ be a graph whose vertices are measurement contexts (orthonormal bases) and whose edges carry observed transition-probability matrices B_e . We distinguish four increasingly demanding notions of admissibility over a field $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$:

vertex data	a context, prior to any transition data
edge admissibility	B_e individually lies in the $\mathbb{F}$ -admissible set
path composition	sequential lifts compose (Chapter 4)
cycle consistency	lifts around a closed loop compose to the identity
global realizability	a single assignment of frames realizes every edge at once

Definition 5.1 (Edgewise vs. global $\mathbb{F}$ -admissibility). Write $\mathcal{A}_{\mathbb{F}}(e)$ for the set of edges individually admitting an $\mathbb{F}$ -lift, and $\mathcal{A}_{\mathbb{F}}(G)$ for the set of context graphs admitting a single assignment of frames to vertices realizing every edge simultaneously.

5.2 Exact Closure and Its Failure Under Projection

For exact frame-to-frame transformations L_{AB} around a triangle of contexts Z, X, Y ,

$$L_{ZX}L_{XY}L_{YZ} = I. \quad (5.1)$$

Applying the probability projection $\pi(L) = |L|^{\circ 2}$ destroys the information needed to express the corresponding closure condition: there is in general no relation

$$B_{ZX}B_{XY}B_{YZ} = I \quad (5.2)$$

satisfied by the observed transition matrices alone.

Ledger 5.2. Equation (5.1) projecting to no corresponding closure relation on the observed B ’s is the central fact of this chapter: exact composition has an identity that probability-level projection does not preserve.

5.3 Gauge Bookkeeping for a Single Real Frame

Before turning to cycles and global consistency, it is worth asking a more basic question about a single edge: recall $\theta \in [0, \pi/4]$ from Chapter 3, the canonical unsigned transition angle between two real two-dimensional contexts. Two very different things compress a raw frame angle α down to this domain, and confusing them would understate what π (the probability projection) actually costs: one is harmless relabeling, the other is a genuine, single-edge loss of information that happens before any cycle or composition enters the picture. Separating them precisely is the point of this section.

Proposition 5.3 (Gauge versus projection loss in the fundamental domain). The canonical unsigned transition angle $\theta \in [0, \pi/4]$ arises from composing two identifications of different character, not one redundancy in disguise:

- (i) Gauge (label swap). $\alpha \sim \alpha + \pi/2$. This identifies the same pair of lines with the labels "+" and "-" exchanged: explicitly $e_+(\alpha + \pi/2) = e_-(\alpha)$ up to sign. Nothing physical changes; only $B_{11} \mapsto 1 - B_{11}$.
- (ii) Projection loss, not gauge. $\alpha \sim \pi - \alpha$. Here $e_+(\pi - \alpha) \neq \pm e_+(\alpha)$ for generic α : these are genuinely different lines. Yet $B_{11}(\alpha) = B_{11}(\pi - \alpha)$ identically, since $\cos^2 \alpha = \cos^2(\pi - \alpha)$. The map π destroys the distinction between these two physically distinct contexts already at the level of a single edge, prior to any composition or cycle.

Verification. Take $\alpha = \pi/6$. Direct computation gives $B_{11}(\pi/6) = 3/4$, $B_{11}(\pi/6 + \pi/2) = 1/4$ (confirming (i), $t \mapsto 1 - t$), and $B_{11}(\pi - \pi/6) = 3/4$ while $e_+(\pi - \pi/6) \neq \pm e_+(\pi/6)$ (confirming (ii): identical observable, distinct context). Composing (i) and (ii) generates the four generic images $\{\alpha, \pi/2 + \alpha, \pi - \alpha, \pi/2 - \alpha\}$ of a representative α , an order-4 group consistent with the $[0, \pi/4]$ fundamental domain. $\square$

Remark 5.4. This confirms the framing above precisely: part (ii) is a single-edge loss of information under π , independent of anything about cycles or global consistency. The cycle-consistency obstruction of §6 is therefore an additional, qualitatively different failure (inability to glue) stacked on top of this more basic single-edge failure (inability to invert π). The two should not be conflated: the $\pm$ signs in the triangle-gluing relation (Theorem 6.2) track both ambiguities at once, which is why four sign combinations appear rather than two.

5.4 Global Gauge for a Context Graph

A gauge decision at one vertex acts on every edge incident to it: for allowed representative changes g_A at each vertex,

$$L_{AB} \mapsto g_A^{-1} L_{AB} g_B. \quad (5.3)$$

Consequently a triangle of contexts is not three independent two-basis problems; signs cannot be chosen edge by edge. Fixing $\alpha_Z = 0$, $\alpha_X = \alpha$, $\alpha_Y = \beta$, the exact relative angles obey

$$\Delta_{ZX} + \Delta_{XY} + \Delta_{YZ} = 0 \pmod{2\pi}, \quad (5.4)$$

which projects, after probability observation $\Delta \mapsto \cos^2 \Delta$, to the weaker triangle-gluing relation of the next section.

5.5 Spanning Trees, Cycle Rank, and General Realizability

The triangle of Chapter 6 is the smallest instance of a general phenomenon, worth isolating for its own sake before specializing back to $n = 3$ contexts.

Definition 5.5 (Context graph). A context graph is a connected graph $G = (V, E)$ whose vertices are measurement contexts and whose edges $e = (A, B)$ each carry an observed bistochastic matrix B_e . Its cycle rank (first Betti number) is

$$\beta_1(G) = |E| - |V| + 1. \quad (5.5)$$

A spanning tree $T \subseteq E$ has $|T| = |V| - 1$ edges and $\beta_1(T) = 0$; each additional edge beyond a spanning tree closes exactly one independent cycle, so $\beta_1(G)$ counts the non-tree edges of any fixed spanning tree.

Proposition 5.6 (Trees carry no independent constraint). Let $T = (V, E_T)$ be a tree, each edge individually $\mathbb{F}$ -admissible (some lift exists reproducing B_e). Then a global frame assignment reproducing every edge of T simultaneously always exists.

Proof. Fix a root $r \in V$ and $F_r := I$. T has a unique path from r to any other vertex v ; choose, for each tree edge, any one admissible lift, and propagate $F_v := F_u L_{uv}$ along that unique path. Since the path to each vertex is unique (this is exactly what makes T a tree), no two propagation routes can disagree, and by construction $F_A^{-1} F_B = L_{AB}$ reproduces B_e on every tree edge. $\square$

Theorem 5.7 (Graph realizability criterion). Let $G = (V, E)$ be a context graph and T a spanning tree. G is globally $\mathbb{F}$ -realizable (some assignment of vertex frames reproduces every edge of E) if and only if there is a choice of admissible lifts on the tree edges of T such that the resulting frame assignment (Proposition 5.6) also reproduces B_e on every non-tree edge $e \in E \setminus T$.

Equivalently: global realizability holds iff there exists a single compatible choice of lifts on a spanning tree for which every non-tree edge simultaneously satisfies its induced fundamental-cycle closure condition. The existential quantifier ranges over one shared tree-lift choice, not over each non-tree edge separately — this is the content of the theorem, not a footnote to it.

Proof. ($\Leftarrow$) The constructed frame assignment reproduces every tree edge by Proposition 5.6 and every non-tree edge by hypothesis, hence every edge of E .

($\Rightarrow$) If some global frame assignment $\{F_A\}$ reproduces every edge of E , its restriction to T is in particular a choice of admissible tree-edge lifts ($L_{AB} := F_A^{-1} F_B$ for $(A, B) \in T$ reproduces B_e by hypothesis), and propagating from that choice reconstructs exactly $\{F_A\}$ (by uniqueness of tree paths), which by hypothesis already matches every non-tree edge. $\square$

The theorem is existential rather than constructive: it characterizes when a compatible assignment exists without prescribing a unique one or a search procedure for finding it. Multiple tree-lift choices can succeed simultaneously (Chapter 6's explicit real solutions to the triangle-gluing relation are one instance), and nothing here selects among them.

Remark 5.8. Theorem 5.7 reduces global realizability to exactly $\beta_1(G)$ simultaneous consistency checks — one per non-tree edge, all against the same tree-lift choice, not checked independently of one another. This is a genuine constraint when $\beta_1(G) \geq 2$: passing each non-tree check in isolation, for possibly different tree-lift choices, does not suffice: it must be a single choice of tree lifts that satisfies every non-tree edge at once.

Corollary 5.9. $\beta_1(K_3) = 3 - 3 + 1 = 1$: the triangle of Chapter 6 is the minimal context graph possessing a nontrivial cycle constraint, and Theorem 6.2 is the $\beta_1 = 1$ case of Theorem 5.7 (spanning tree $\{ZX, ZY\}$, single non-tree edge XY).

Remark 5.10 (Two independent axes, not one ladder). Theorem 5.7 should not be read as one more rung on a single vertex $\rightarrow$ edge $\rightarrow$ path $\rightarrow$ cycle $\rightarrow$ global hierarchy. Part I actually establishes two logically independent questions, and collapsing them into one ordered scale obscures that independence.

Compositional sufficiency (Chapter 4) asks whether projected data retains enough information to predict sequential composition: $\pi(L_2 L_1) \stackrel{?}{=} \pi(L_2) \pi(L_1)$. This question is already meaningful on a bare open path — no cycle, and in particular no closed loop in a context graph, is required to ask or answer it. It measures information lost by projection, full stop.

Global realizability (this chapter and Chapter 6) asks whether independently specified, individually admissible edge data $\{B_e\}$ on a context graph admits one simultaneous vertex-frame assignment reproducing all of it at once. This is where trees, non-tree edges, and $\beta_1(G)$ belong, and it is silent on whether any particular composition of chosen lifts is faithful.

The MUB triangle fails the second test; the interference calculation of Chapter 4 demonstrates failure of the first. Nothing forces these into a common scale. An assumption's contribution to the reconstruction is better recorded as a pair

$$Q(A) = (Q_{\text{comp}}(A), Q_{\text{glob}}(A)) \quad (5.6)$$

— what it contributes to compositional recovery, and separately what it contributes to global realizability — rather than a single numerical "depth." Part II introduces a third, independent axis in the same spirit: system composition (does an individually admissible state space for each of several systems compose into an admissible joint state space?), which is neither sequential composition along a path nor contextual gluing around a cycle.

Chapter 6

The Triangle-Gluing Theorem and the Real MUB Obstruction

6.1 Exact Closure Upstairs

Represent each of the three contexts Z, X, Y by a real angle: fix $a_Z = 0$ by an overall choice of coordinates (always available, and immaterial to any observable transition matrix), and write $a_X = \alpha$, $a_Y = \beta$ for real numbers. With $L_{AB} := R(a_B - a_A)$ as in Chapter 5,

$$L_{ZX}L_{XY}L_{YZ} = R(\alpha)R(\beta - \alpha)R(-\beta) = R(0) = I \quad (6.1)$$

exactly, with no quotient or approximation: this is the additive law for rotation angles, true for every choice of $\alpha, \beta \in \mathbb{R}$.

6.2 The Fibre of the Probability Projection

For a labeled real angle a , the diagonal transition probability relative to the fixed reference at angle 0 is $t(a) = \cos^2 a$, and the canonical unsigned observable is $\theta(a) := \min(\arccos \sqrt{t(a)}, \pi/2 - \arccos \sqrt{t(a)}) \in [0, \pi/4]$, invariant under the label-swap gauge of §5.3. Restrict attention to $\alpha, \beta \in [0, \pi/2]$, where $\arccos \sqrt{\cos^2 \theta} = \theta$ identically, so $\theta(\alpha) = \alpha$ before folding and $\theta_{ZX} := \theta(\alpha)$, $\theta_{ZY} := \theta(\beta)$ may be taken as the raw angles themselves on this range.

Lemma 6.1 (Closed-form composition identity). For $\theta_1, \theta_2 \in [0, \pi/2]$,

$$1 - \cos^2 \theta_1 - \cos^2 \theta_2 + 2 \cos^2 \theta_1 \cos^2 \theta_2 \pm 2 \sqrt{\cos^2 \theta_1 \cos^2 \theta_2 \sin^2 \theta_1 \sin^2 \theta_2} = \cos^2(\theta_1 \mp \theta_2). \quad (6.2)$$

Proof. Expand $\cos^2(\theta_1 - \theta_2) = (\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2)^2 = \cos^2 \theta_1 \cos^2 \theta_2 + \sin^2 \theta_1 \sin^2 \theta_2 + 2 \sin \theta_1 \cos \theta_1 \sin \theta_2 \cos \theta_2$. Since $\theta_1, \theta_2 \in [0, \pi/2]$, $\sin \theta_i, \cos \theta_i \geq 0$, so $\sin \theta_i \cos \theta_i = \sqrt{\cos^2 \theta_i \sin^2 \theta_i}$, and $\sin^2 \theta_1 \sin^2 \theta_2 = (1 - \cos^2 \theta_1)(1 - \cos^2 \theta_2)$; expanding the latter and collecting terms gives the stated identity with the $+$ sign, matching $\cos^2(\theta_1 - \theta_2)$. The $-$ sign case follows identically using $\cos^2(\theta_1 + \theta_2) = (\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2)^2$. $\square$

Writing $t_i = \cos^2 \theta_i$, Lemma 6.1 says exactly that $\cos^2(\beta - \alpha)$, computed from $t_{ZX} = \cos^2 \alpha$ and $t_{ZY} = \cos^2 \beta$ alone, is not single-valued: it is one of the two roots

$$t_{XY} \in \{ \cos^2(\theta_{ZX} - \theta_{ZY}), \cos^2(\theta_{ZX} + \theta_{ZY}) \} \quad (6.3)$$

according to the (unobservable) relative sign of $\sin \alpha$ and $\sin \beta$. This is the precise fibre statement in place of the earlier proof sketch: the projection loses exactly one bit — a relative sign — beyond the diagonal probabilities t_{ZX}, t_{ZY} themselves.

6.3 Descent: the Triangle-Gluing Theorem

Theorem 6.2 (Triangle gluing in $\mathbb{R}^2$). Let $\theta_{ZX}, \theta_{ZY}, \theta_{XY} \in [0, \pi/4]$ be the canonical unsigned transition angles of three pairwise 2×2 bistochastic matrices on contexts Z, X, Y . A real assignment of frames to Z, X, Y jointly realizing all three exists if and only if

$$\theta_{XY} \equiv \pm\theta_{ZX} \pm \theta_{ZY} \pmod{\pi/2} \quad (6.4)$$

for some choice of signs.

Proof. ($\Rightarrow$, necessity.) Immediate from (6.1) and Lemma 6.1: any real realization has θ_{XY} equal, before its own canonical folding, to $\theta_{ZX} \pm \theta_{ZY}$ for one of the two signs; folding to $[0, \pi/4]$ turns equality into the congruence (6.4).

($\Leftarrow$, sufficiency.) Suppose (6.4) holds for some choice of signs. Fix $a_Z = 0$, $a_X = \theta_{ZX}$, and

$$a_Y = \begin{cases} \theta_{ZY} & \text{if the relation holds with } \theta_{XY} \equiv \theta_{ZX} - \theta_{ZY} \pmod{\pi/2}, \\ -\theta_{ZY} & \text{if it holds with } \theta_{XY} \equiv \theta_{ZX} + \theta_{ZY} \pmod{\pi/2}. \end{cases} \quad (6.5)$$

By construction $\cos^2 a_X = \cos^2 \theta_{ZX}$ and $\cos^2 a_Y = \cos^2(\pm\theta_{ZY}) = \cos^2 \theta_{ZY}$, reproducing B_{ZX} and B_{ZY} exactly; and $a_Y - a_X$ equals $-(\theta_{ZX} + \theta_{ZY})$ or $\theta_{ZY} - \theta_{ZX}$ respectively, whose canonical fold is θ_{XY} by hypothesis, reproducing B_{XY} . This exhibits an explicit real frame assignment realizing all three matrices. $\square$

Remark 6.3. This theorem is the general statement; mutual unbiasedness is the sharpest special case, not the only one. But "sharpest" should not be read as "rare": for independently and continuously distributed edge angles, realizability itself is the exceptional condition. Fixing any two of the three angles, Theorem 6.2 shows the third is constrained to one of at most two specific values out of the continuum $[0, \pi/4]$ — a codimension-one condition. If all three angles are drawn independently from a continuous distribution, the probability of hitting that condition exactly is zero. Realizability is not generic; it is the non-generic, measure-zero case, true for every graph containing a cycle, not a special feature of the MUB point. What is special about mutual unbiasedness is symmetry, not scarcity: it is the unique maximally symmetric point at which the (already generic) failure occurs, not the rare exception to an otherwise generic success.

6.4 The Mutually Unbiased Corollary

Corollary 6.4 (Real MUB obstruction, $d = 2$). There do not exist three pairwise mutually unbiased orthonormal bases of $\mathbb{R}^2$.

Proof. Set $\theta_{ZX} = \theta_{ZY} = \theta_{XY} = \pi/4$ (mutual unbiasedness). Theorem 6.2 requires $\pi/4 \equiv \pm\pi/4 \pm \pi/4 \pmod{\pi/2}$. The right-hand side takes only the values 0 or $\pi/2$, both $\equiv 0 \pmod{\pi/2}$, never $\pi/4$. Contradiction. (Equivalently, via Lemma 6.1 directly: with $\theta_1 = \theta_2 = \pi/4$, $\cos^2(\theta_1 \mp \theta_2) \in \{\cos^2 0, \cos^2(\pi/2)\} = \{1, 0\}$, never $\cos^2(\pi/4) = 1/2$.) $\square$

Scope. This rules out $\mathbb{R}^2$ for one specific configuration — three mutually unbiased contexts — not for real-valued physics in general. It does not show $\mathbb{R}$ fails whenever a cycle is present (most cycle data is real-realizable; §6.4's remark below makes this explicit), and it does not rule out an enlarged real state space or a quaternionic alternative, both explicitly untouched here (this chapter's own closing ledger says so). Treat "real numbers cannot do this" as scoped to this exact triangle, this exact dimension, this exact data — not as a

general verdict on $\mathbb{R}$.

Remark 6.5. Corollary 6.4 is the $d = 2$ specialization of the general real-MUB bound $N_{\mathbb{R}}(d) \leq \lfloor d/2 \rfloor + 1$ of Boykin, Sitharam, Tarifi, and Wocjan [4]. The bound is not new; what the reconstruction contributes is the interpretation: every edge of the context triangle is individually orthostochastically admissible, yet the triangle as a whole has no realization in $\mathbb{R}^2$, while $\mathbb{C}^2$ realizes all three simultaneously via

$$Z = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}, \quad X = \frac{1}{\sqrt{2}} \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}, \quad Y = \frac{1}{\sqrt{2}} \left\{ \begin{pmatrix} 1 \\ i \end{pmatrix}, \begin{pmatrix} 1 \\ -i \end{pmatrix} \right\}. \quad (6.6)$$

Ledger 6.6 (End of Part I, single-system results). Established: edgewise real admissibility does not imply global real admissibility (Theorem 6.2, proved in both directions: necessity from the closed-form fibre identity of Lemma 6.1, sufficiency by explicit construction); the mutually unbiased triple is the maximally inconsistent case (Corollary 6.4); this rules out $\mathbb{R}^2$ as a global lift for three pairwise-MUB contexts without invoking composite systems.

Not established: that quantum mechanics must be complex in general; enlarged real state spaces and quaternionic alternatives remain untouched by this argument. The uniqueness question is deferred to Part II (composite systems, local tomography).

Remark 6.7 (A connection noticed, not pursued). The cycle-based obstruction of this chapter has the flavor of an early, primitive version of holonomy — a closed loop carrying information no single edge does, the same shape as Berry phase or a $\mathbb{Z}_2$ -valued cocycle failing to trivialize. This book does not develop that connection: Part IV pursued higher-dimensional and quaternionic witnesses instead of a Berry-phase chapter, so the analogy above should be read as exactly that — an analogy noticed in passing, not a bundle/cocycle formulation worked out rigorously anywhere in this book. It is recorded here as a genuine open direction rather than silently dropped.

Part II

Composite Systems and the Scalar Field

Chapter 7

Minimal Witnesses and the Extremal Reformulation

Every chapter so far has deliberately stayed within a single system, observed through different contexts or composed sequentially with itself — no two separate systems have appeared together anywhere in Part I. That was not an oversight; it isolates what a single system’s admissibility structure can and cannot decide before asking the different, genuinely new question this part takes up: what happens when independently prepared systems are combined? The assumptions introduced from here on (Chapter 8’s local tomography, most prominently) answer that new question — they do not patch or strengthen anything left unresolved in Part I.

Part I answers, for a specific example, the question of whether edgewise real-admissible data can fail to be globally real-realizable. This chapter reformulates that question in extremal terms: not merely whether a failure occurs, but what the smallest structure exhibiting it looks like — and uses that reformulation to state precisely what Part II must still establish.

7.1 Realizability Spaces

Definition 7.1 (Realizability space). Fix a context graph $G = (V, E)$, a local dimension d (each vertex a d -outcome context), and a scalar field $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$. Let $\mathcal{D}(G, d)$ be the space of edge-labelings $\{B_e\}_{e \in E}$ by $d \times d$ bistochastic matrices. The realizability space

$$\mathcal{R}_{\mathbb{F}}(G, d) \subseteq \mathcal{D}(G, d) \tag{7.1}$$

is the subset of edge-labelings admitting a global assignment of vertex frames over $\mathbb{F}$ reproducing every B_e simultaneously (Theorem 5.7).

Since $\mathbb{R} \subset \mathbb{C} \subset \mathbb{H}$ as scalar extensions with matching inclusions of admissible lifts, $\mathcal{R}_{\mathbb{R}}(G, d) \subseteq \mathcal{R}_{\mathbb{C}}(G, d) \subseteq \mathcal{R}_{\mathbb{H}}(G, d)$ for every G, d .

7.2 Minimal Witnesses

Definition 7.2 (Minimal witness). For fields $\mathbb{F} \subsetneq \mathbb{F}'$ and dimension d , a witness is a context graph G with $\mathcal{R}_{\mathbb{F}}(G, d) \subsetneq \mathcal{R}_{\mathbb{F}'}(G, d)$. Define

$$m_{\beta}(\mathbb{F}, \mathbb{F}'; d) := \min\{\beta_1(G) : G \text{ is a witness}\}. \tag{7.2}$$

Theorem 7.3 (The $d = 2$ real/complex minimal witness). $m_{\beta}(\mathbb{R}, \mathbb{C}; 2) = 1$, achieved by $G = K_3$.

Proof. $\beta_1(G) = 0$ (any tree) gives $\mathcal{R}_{\mathbb{R}}(T, 2) = \mathcal{R}_{\mathbb{C}}(T, 2)$ directly from Proposition 5.6: any individually admissible edge data on a tree is realizable over $\mathbb{R}$ (hence trivially over $\mathbb{C}$ too), so

no tree separates the two fields, and $m_\beta \geq 1$. The mutually-unbiased edge-labeling of Chapter 6 lies in $\mathcal{R}_\mathbb{C}(K_3, 2) \setminus \mathcal{R}_\mathbb{R}(K_3, 2)$ by Corollary 6.4, so K_3 (with $\beta_1(K_3) = 1$) is a witness, giving $m_\beta(\mathbb{R}, \mathbb{C}; 2) \leq 1$. $\square$

Remark 7.4. This is exactly Corollary 5.9 restated in extremal language: the triangle is not merely a convenient example, it is provably the smallest possible one, in the cycle-rank sense, for $d = 2$. Theorem 7.3 is what Part I actually establishes; the questions below are what it does not establish, and are stated as open directions rather than results.

7.3 Open Extremal Questions

The following questions are natural given Definitions 7.1–7.2, but nothing proved so far in this book answers them. They are recorded here as a research programme, not as claims.

Higher-dimensional minimal witnesses. What is $m_\beta(\mathbb{R}, \mathbb{C}; d)$ for $d \geq 3$? Chapter 3’s $n \geq 3$ incompleteness theorem gives a single-edge (β_1 undefined, since it concerns one bistochastic matrix rather than a graph) separation at $d = 3$ via the DFT example; whether this induces a graph-level minimal witness smaller than, equal to, or larger than the $d = 2$ value 1 is open.

Quaternionic separation. $m_\beta(\mathbb{C}, \mathbb{H}; d)$ is entirely unaddressed by anything proved in this book. The composite-system chapters that follow are the natural place to make first contact with it, but no claim is made here about where or whether such a witness exists at small β_1 .

Forbidden-subgraph characterization. Is there a finite family $\mathcal{F}$ of forbidden context subgraphs such that $\mathcal{R}_\mathbb{R}(G, d) = \mathcal{D}(G, d)$ (no separation at all) if and only if G contains no member of $\mathcal{F}$? This is a genuine open combinatorial question; nothing here establishes even that such a finite family exists.

Density thresholds. Given a random context graph model, does the probability that a uniformly random admissible edge-labeling requires $\mathbb{C}$ (or $\mathbb{H}$) rather than $\mathbb{R}$ approach 1 as edge density increases, in the manner of an Erdős–Rényi threshold? This requires specifying a probability measure on $\mathcal{D}(G, d)$ that has not been defined anywhere in this book, and the analogy to random-graph phase transitions should not be overstated: an Erdős–Rényi threshold concerns the asymptotic graph structure as edge probability varies, whereas this question, if posed precisely, would concern a measure on the edge-label data on a fixed or growing graph — a different (if analogous) kind of threshold question, not a restatement of the classical one.

Ledger 7.5 (Opening of Part II). Established: the realizability-space and minimal-witness vocabulary (Definitions 7.1–7.2) is well-posed, and $m_\beta(\mathbb{R}, \mathbb{C}; 2) = 1$ exactly, with K_3 as the minimal witness (Theorem 7.3), reformulating Part I’s central result in extremal terms without adding new mathematical content.

Not established: every question in §7.3 — higher- d minimal witnesses, any quaternionic separation, forbidden-subgraph characterizations, and density thresholds all remain open. Part II’s composite-system chapters address a related but distinct question (system composition, the third axis of the book’s architecture) and should not be read as resolving these extremal questions in passing.

Chapter 8

Composite Systems and Local Tomography

Part I's two axes — compositional sufficiency (Chapter 4) and global realizability (Chapters 5–6) — both concern a single system observed through different contexts or composed sequentially with itself. This chapter opens the third axis: given two separate systems, individually admissible in the sense of Part I, how must their joint description compose?

8.1 The Composition Problem

Let systems A, B have individual state spaces of dimension d_A, d_B respectively (each already subject to the admissibility constraints of Chapters 1–3). Operational independence means: A and B can be prepared and measured separately, and a joint preparation of both is a legitimate object of study without presupposing in advance what mathematical structure describes it. The question this chapter asks is deliberately neutral:

$$\boxed{\text{How must independently accessible systems compose?}} \quad (8.1)$$

rather than assuming a complex tensor product from the outset and asking only whether it is consistent.

8.2 $\mathbb{F}$ -Hermitian State Spaces

For each candidate scalar field $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$, a natural (not yet composite) state space for a d -dimensional system is the space of $\mathbb{F}$ -Hermitian $d \times d$ matrices ρ (density operators $\rho \geq 0$, $\text{tr } \rho = 1$ living inside this linear space), where $\mathbb{F}$ -Hermitian means $\rho^\dagger = \rho$ with the conjugate taken in $\mathbb{F}$.

Proposition 8.1 (Dimension of the Hermitian state space). The real vector space of $\mathbb{F}$ -Hermitian $d \times d$ matrices has dimension

$$\beta_{\mathbb{R}}(d) = \frac{d(d+1)}{2}, \quad \beta_{\mathbb{C}}(d) = d^2, \quad \beta_{\mathbb{H}}(d) = d(2d-1). \quad (8.2)$$

Proof. In each case, Hermiticity forces the diagonal entries to be real (a self-conjugate element of $\mathbb{R}, \mathbb{C}, \mathbb{H}$ is real), contributing d real parameters, while each of the $d(d-1)/2$ upper-triangular off-diagonal entries is an unconstrained element of $\mathbb{F}$ (the lower triangle being determined by conjugating the upper), contributing $\dim_{\mathbb{R}}(\mathbb{F})$ real parameters each: 1 for $\mathbb{R}$, 2 for $\mathbb{C}$, 4 for $\mathbb{H}$. Summing,

$$\beta_{\mathbb{R}}(d) = d + 1 \cdot \frac{d(d-1)}{2} = \frac{d(d+1)}{2}, \quad \beta_{\mathbb{C}}(d) = d + 2 \cdot \frac{d(d-1)}{2} = d^2, \quad \beta_{\mathbb{H}}(d) = d + 4 \cdot \frac{d(d-1)}{2} = d(2d-1). \quad (8.3)$$

□

8.3 Local Tomography

Definition 8.2 (Local tomography, dimension-counting form). A scalar field $\mathbb{F}$ satisfies local tomography if, for every d_A, d_B , the Hermitian state space of the composite system has dimension equal to the product of the local dimensions:

$$\beta_{\mathbb{F}}(d_A d_B) = \beta_{\mathbb{F}}(d_A) \beta_{\mathbb{F}}(d_B). \quad (8.4)$$

Informally: the joint state is exactly as informative as a tomographically complete set of local measurements on each subsystem, with no additional holistic parameters and no missing ones.

Theorem 8.3 (Local tomography selects $\mathbb{C}$). For all $d_A, d_B \geq 2$:

$$\beta_{\mathbb{C}}(d_A d_B) - \beta_{\mathbb{C}}(d_A) \beta_{\mathbb{C}}(d_B) = 0, \quad (8.5)$$

$$\beta_{\mathbb{R}}(d_A d_B) - \beta_{\mathbb{R}}(d_A) \beta_{\mathbb{R}}(d_B) = \frac{d_A d_B (d_A - 1)(d_B - 1)}{4} > 0, \quad (8.6)$$

$$\beta_{\mathbb{H}}(d_A d_B) - \beta_{\mathbb{H}}(d_A) \beta_{\mathbb{H}}(d_B) = -2 d_A d_B (d_A - 1)(d_B - 1) < 0. \quad (8.7)$$

Proof. Direct substitution of Proposition 8.1 into each difference, using $d_A d_B - d_A - d_B + 1 = (d_A - 1)(d_B - 1)$:

$$\beta_{\mathbb{C}}(d_A d_B) - \beta_{\mathbb{C}}(d_A) \beta_{\mathbb{C}}(d_B) = (d_A d_B)^2 - d_A^2 d_B^2 = 0, \quad (8.8)$$

$$\beta_{\mathbb{R}}(d_A d_B) - \beta_{\mathbb{R}}(d_A) \beta_{\mathbb{R}}(d_B) = \frac{d_A d_B (d_A d_B + 1)}{2} - \frac{d_A (d_A + 1)}{2} \cdot \frac{d_B (d_B + 1)}{2} = \frac{d_A d_B (d_A - 1)(d_B - 1)}{4}, \quad (8.9)$$

$$\beta_{\mathbb{H}}(d_A d_B) - \beta_{\mathbb{H}}(d_A) \beta_{\mathbb{H}}(d_B) = d_A d_B (2 d_A d_B - 1) - d_A (2 d_A - 1) d_B (2 d_B - 1) = -2 d_A d_B (d_A - 1)(d_B - 1). \quad (8.10)$$

For $d_A, d_B \geq 2$, $(d_A - 1)(d_B - 1) > 0$, giving the stated signs. $\square$

Remark 8.4. The mismatch is not a boundary artifact of small dimensions: it grows without bound as d_A, d_B increase, in opposite directions for $\mathbb{R}$ and $\mathbb{H}$. Real composite systems always carry more joint parameters than local tomography predicts (extra, non-locally-accessible correlational structure); quaternionic composite systems always carry fewer (the naive product construction is already overdetermined, consistent with the well-known difficulty of defining a well-behaved tensor product for quaternionic Hilbert spaces at all). Complex Hermitian state spaces are the unique case among the three where the identity holds exactly, for every pair of dimensions.

Scope. "Unique" above means unique among $\mathbb{R}, \mathbb{C}, \mathbb{H}$, under this specific dimension-counting test, using the ordinary Kronecker product. It is not a claim that $\mathbb{C}$ is the only conceivable scalar structure for physics, and it is not unconditional: local tomography itself is an assumed principle this chapter tests rather than derives (this chapter's closing ledger says so explicitly), and Chapter 18's §18.4 discusses published work in which changing the composition rule (not the scalar field) lets a real theory pass an equivalent test. Read this theorem as "given local tomography and the Kronecker product, $\mathbb{C}$ is the unique fixed point" — every word of that qualification is load-bearing.

Example 8.5 ($d_A = d_B = 2$). Two real "qubits": $\beta_{\mathbb{R}}(4) = 10$ vs. $\beta_{\mathbb{R}}(2)^2 = 9$ — one real parameter of the joint two-qubit state is invisible to any local tomography scheme. Two quaternionic "qubits": $\beta_{\mathbb{H}}(4) = 28$ vs. $\beta_{\mathbb{H}}(2)^2 = 36$ — the naive quaternionic product already overcounts by eight real parameters before any physical constraint is imposed. Two ordinary qubits: $\beta_{\mathbb{C}}(4) = 16 = \beta_{\mathbb{C}}(2)^2$ exactly.

Ledger 8.6 (Scope of this chapter). Established: local tomography, precisely defined as a dimension-counting identity on Hermitian state spaces, holds identically for $\mathbb{C}$ and fails identically (in opposite directions) for $\mathbb{R}$ and $\mathbb{H}$, for every pair of dimensions $d_A, d_B \geq 2$ (Theorem 8.3). This argument is not original to this book; it is the classical parameter-counting characterization due to Araki [6], sharpened by Wootters and later incorporated into Hardy’s axiomatic reconstruction of quantum theory [7].

Not established: that local tomography is itself forced by anything proved in Part I — it is introduced here as an additional principle under test, in keeping with the book’s assumption-ledger methodology, not as a consequence of the single-system results. Nor does dimension-counting alone establish the full tensor-product structure, the Born rule for composite measurements, or entanglement; it constrains the dimension of the composite state space and no more. What Part I’s cycle obstruction (Chapter 6) and this chapter’s local-tomography argument have in common — ruling out $\mathbb{R}$ — arise by entirely different mechanisms (contextual gluing around a cycle vs. parameter counting across a tensor product) and should be credited as independent lines of evidence, not conflated into one argument.

Chapter 9

The Tensor Product as the Forced Composition Rule

Chapter 8 showed that the dimensions of composite and product Hermitian state spaces match exactly for $\mathbb{C}$ and fail to match for $\mathbb{R}, \mathbb{H}$. That leaves open whether the composite operator space for $\mathbb{C}$ is actually spanned by products of local operators — i.e. whether the tensor product construction is not merely dimensionally consistent but the concrete realization of local tomography. This chapter proves that it is, and exhibits explicitly what goes missing for $\mathbb{R}$.

9.1 Local Bases and Their Products

Fix an $\mathbb{R}$ -basis $\{\sigma_i\}_{i=1}^{\beta_{\mathbb{C}}(d)}$ of the space of $d \times d$ complex Hermitian matrices $\text{Herm}_{\mathbb{C}}(d)$ (e.g. the identity together with the generalized Gell-Mann matrices). For a Hermitian X , $X = \frac{1}{2}(X + X^\dagger)$ trivially, so every complex $d \times d$ matrix decomposes uniquely as $H_1 + iH_2$ with $H_1, H_2 \in \text{Herm}_{\mathbb{C}}(d)$; consequently $\{\sigma_i\}$, an $\mathbb{R}$ -basis of $\text{Herm}_{\mathbb{C}}(d)$, is also a $\mathbb{C}$ -basis of the full matrix algebra $M_d(\mathbb{C})$.

Theorem 9.1 (Tensor products of local bases span the composite). Let $\{\sigma_i\}_{i=1}^{d_A^2}$, $\{\tau_j\}_{j=1}^{d_B^2}$ be $\mathbb{R}$ -bases of $\text{Herm}_{\mathbb{C}}(d_A)$, $\text{Herm}_{\mathbb{C}}(d_B)$. Then $\{\sigma_i \otimes \tau_j\}$ is an $\mathbb{R}$ -basis of $\text{Herm}_{\mathbb{C}}(d_A d_B)$.

Proof. Each $\sigma_i \otimes \tau_j$ is Hermitian, since $(\sigma_i \otimes \tau_j)^\dagger = \sigma_i^\dagger \otimes \tau_j^\dagger = \sigma_i \otimes \tau_j$. By the preceding remark, $\{\sigma_i\}$ and $\{\tau_j\}$ are $\mathbb{C}$ -bases of $M_{d_A}(\mathbb{C})$ and $M_{d_B}(\mathbb{C})$ respectively; since $M_{d_A}(\mathbb{C}) \otimes_{\mathbb{C}} M_{d_B}(\mathbb{C}) \cong M_{d_A d_B}(\mathbb{C})$ via the Kronecker product (a standard fact about matrix algebras), $\{\sigma_i \otimes \tau_j\}$ is a $\mathbb{C}$ -basis of $M_{d_A d_B}(\mathbb{C})$: in particular $\mathbb{C}$ -linearly independent, hence also $\mathbb{R}$ -linearly independent (an $\mathbb{R}$ -linear dependence is a special case of a $\mathbb{C}$ -linear one). This gives $d_A^2 d_B^2 = \beta_{\mathbb{C}}(d_A d_B)$ (Theorem 8.3) $\mathbb{R}$ -linearly independent Hermitian elements inside the real vector space $\text{Herm}_{\mathbb{C}}(d_A d_B)$, which has exactly that real dimension; they therefore form a basis. $\square$

Remark 9.2. This upgrades Chapter 8's dimension count to a concrete claim: for $\mathbb{C}$, every composite observable is a real-linear combination of tensor products of local observables. Local tomography for $\mathbb{C}$ is not a coincidence of arithmetic; the tensor product construction is its literal witness.

9.2 What Goes Missing for $\mathbb{R}$: an Explicit Example

The same construction over $\mathbb{R}$ shows exactly where local tomography fails. Take $d_A = d_B = 2$. A real basis of $\text{Herm}_{\mathbb{R}}(2)$ (real symmetric 2×2 matrices) is $\{I, X, Z\}$ with $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

— only $3 = \beta_{\mathbb{R}}(2)$ elements, since the fourth generator needed to span all of $M_2(\mathbb{R})$,

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad (9.1)$$

is antisymmetric ($J^\top = -J$), not a valid local observable over $\mathbb{R}$. (Chapter 19 gives J 's full elementary derivation as the generator of quarter-turns and shows it is, identically, the real matrix representing i throughout this book and in the recent literature discussed in Chapter 18.)

Remark 9.3. J should not be confused with the complex Pauli matrix $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, which is Hermitian over $\mathbb{C}$ and belongs to $\text{Herm}_{\mathbb{C}}(2)$. J is the real, scalar-stripped matrix $-i\sigma_y$; it is a perfectly good real antisymmetric matrix, just not a real symmetric one, and therefore not itself an admissible local observable in the real theory of Chapter 8.

Proposition 9.4. The nine products $\{I, X, Z\} \otimes \{I, X, Z\}$ are linearly independent elements of $\text{Herm}_{\mathbb{R}}(4)$, but do not span it: $J \otimes J$ is symmetric,

$$(J \otimes J)^\top = J^\top \otimes J^\top = (-J) \otimes (-J) = J \otimes J, \quad (9.2)$$

hence lies in $\text{Herm}_{\mathbb{R}}(4)$, and is linearly independent of the nine products above.

Proof. Linear independence of the nine products, and of $J \otimes J$ from them, follows because $\{I, X, Z, J\}$ is a full basis of $M_2(\mathbb{R})$ (dimension $4 = 2^2$), so $\{I, X, Z, J\} \otimes \{I, X, Z, J\}$ (sixteen elements) is a basis of $M_4(\mathbb{R})$ (dimension $16 = 4^2$) by the same Kronecker-product-of-bases fact used in Theorem 9.1; $J \otimes J$ is one of these sixteen basis elements, hence independent of the other fifteen, in particular the nine drawn only from $\{I, X, Z\} \otimes \{I, X, Z\}$. $\square$

Remark 9.5. This is the exact real tenth dimension counted abstractly in Theorem 8.3 ($\beta_{\mathbb{R}}(4) = 10$ vs. $\beta_{\mathbb{R}}(2)^2 = 9$), now identified concretely. J is not a locally realizable real observable — it is antisymmetric — yet $J \otimes J$ is a legitimate real symmetric observable on the composite system, invisible to any measurement scheme built only from products of local real observables. Real composite systems carry a genuinely nonlocal degree of freedom built from two individually-inadmissible local generators whose product happens to be admissible; this is the concrete mechanism behind Chapter 8's abstract dimension mismatch, not merely a restatement of it.

Remark 9.6 (Quaternionic overcompleteness). The mirror phenomenon occurs for $\mathbb{H}$, in the opposite direction: the analogous product basis is overcomplete ($\beta_{\mathbb{H}}(d_A)\beta_{\mathbb{H}}(d_B) > \beta_{\mathbb{H}}(d_A d_B)$ for $d_A, d_B \geq 2$, Theorem 8.3), so the products $\{\sigma_i \otimes \tau_j\}$ must satisfy nontrivial linear relations rather than failing to span. This is consistent with the well-documented difficulty of defining a well-behaved, associative tensor product for quaternionic Hilbert spaces [8]; this book does not attempt to resolve that difficulty, only to note that it is exactly where the dimension count predicts trouble.

Ledger 9.7 (End of Chapter 9). Established: for $\mathbb{C}$, tensor products of local Hermitian bases do not merely match the composite dimension but literally span the composite Hermitian operator space (Theorem 9.1); for $\mathbb{R}$, the missing dimension is identified concretely as spanned by $J \otimes J$, a product of two locally-inadmissible generators (Proposition 9.4).

Not established: the positivity cone, normalization, or Born-rule structure of the composite state space — this chapter concerns only the ambient linear space of Hermitian operators, which observables and states must live inside but does not by itself determine which elements are valid (positive, normalized) states. Entanglement, the tensor-product Born rule $p_i = \text{tr}(\rho E_i)$ for composite measurements, and the partial trace remain to be established in later chapters of the recovery layer.

Chapter 10

Positivity and the State Space

Chapter 9 characterized the ambient linear space of Hermitian operators a composite system's observables must live in. Not every element of that space is a physically admissible state: this chapter derives the further restriction to positive, normalized operators, motivated operationally rather than posited.

10.1 Positivity as an Admissibility Condition

An operator $\rho \in \text{Herm}_{\mathbb{C}}(d)$ is proposed as a description of a system's state via the rule $p = \text{tr}(\rho P)$ for a measurement projector P . Admissibility requires this to be a genuine probability for every rank-one projector $P_\psi = |\psi\rangle\langle\psi|$ corresponding to a legitimate test:

$$\text{tr}(\rho P_\psi) = \langle\psi|\rho|\psi\rangle \geq 0 \quad \forall |\psi\rangle. \quad (10.1)$$

This is exactly the definition of positive semidefiniteness, $\rho \succeq 0$. Total-probability normalization over any complete measurement (any orthonormal basis) forces $\text{tr} \rho = 1$. Together,

$$\mathcal{D}_d(\mathbb{C}) = \{\rho \in \text{Herm}_{\mathbb{C}}(d) : \rho \succeq 0, \text{tr} \rho = 1\}. \quad (10.2)$$

Remark 10.1. Positivity is not introduced here as an additional quantum postulate; it is forced by the same admissibility logic used throughout the book — every proposed observable-probability pairing must actually be a probability — and sits inside the ambient linear space exactly as $\text{DS}(n)$ sits inside its ambient matrix space in Chapter 2, or $\text{US}(n)$ inside $\text{DS}(n)$ in Chapter 3: a further admissibility cut on a previously characterized ambient space, not a new kind of assumption.

10.2 The Real Composite Direction, Revisited

Chapter 9 identified $J \otimes J$ as the real-composite direction invisible to any purely local (product-basis) description. Now that positivity is available, this direction can be asked a sharper question: not merely is it in the ambient space, but for which coefficients is it an actual state?

Example 10.2 (The bound on the invisible direction). Consider the one-parameter family on two qubits,

$$\rho(c) = \frac{1}{4}(I \otimes I + c J \otimes J), \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \quad (10.3)$$

Since $J^2 = -I$, $(J \otimes J)^2 = J^2 \otimes J^2 = (-I) \otimes (-I) = I$, so $J \otimes J$ has eigenvalues ± 1 (each with multiplicity 2, as $J \otimes J$ is a 4×4 traceless involution). The eigenvalues of $\rho(c)$ are therefore $\frac{1}{4}(1 \pm c)$, each doubly degenerate. Positivity requires both signs nonnegative:

$$\rho(c) \succeq 0 \iff |c| \leq 1. \quad (10.4)$$

$\text{tr} \rho(c) = 1$ automatically, since $\text{tr}(J \otimes J) = 0$.

Remark 10.3. This is worth pausing on. Chapter 9 found $J \otimes J$ as an extra basis vector — a fact about dimension counting. Positivity turns it into a bounded physical degree of freedom: c ranges over a genuine interval $[-1, 1]$ of admissible states, not an unconstrained coordinate. The dimension mismatch of Chapter 8 was never merely formal; it names an axis along which real composite systems can actually vary their state, continuously, within the constraints derived in this chapter.

Ledger 10.4 (End of Chapter 10). Established: positivity is derived from the requirement that every proposed rank-one test give a nonnegative probability, not posited separately; $\mathcal{D}_d(\mathbb{C})$ is characterized as positive, trace-one elements of the ambient space of Chapter 9; the locally-invisible real direction $J \otimes J$ is shown to support a genuine one-parameter family of admissible states, $|c| \leq 1$, with the bound derived from $(J \otimes J)^2 = I$.

Not established: anything about what happens to this direction under marginalization (Chapter 11), or about entanglement and separability of $\rho(c)$, which require the partial trace machinery not yet introduced.

Chapter 11

Partial Trace and Marginal Invisibility

11.1 The Partial Trace

For a composite system AB with state $\rho_{AB} \in \mathcal{D}_{d_A d_B}(\mathbb{C})$, the reduced (marginal) state of subsystem A is defined by the partial trace over B : writing ρ_{AB} in block form indexed by A 's basis, with blocks $(\rho_{AB})_{ij}$ acting on B 's Hilbert space,

$$(\rho_A)_{ij} = \text{tr}_B(\rho_{AB})_{ij} := \text{tr}[(\rho_{AB})_{ij}], \quad (11.1)$$

the ordinary trace of each block. ρ_A is exactly the operator whose predictions match ρ_{AB} on every measurement performed on A alone (with B un-measured): for any local operator X_A on A ,

$$\text{tr}_A(\rho_A X_A) = \text{tr}_{AB}(\rho_{AB}(X_A \otimes I_B)). \quad (11.2)$$

The partial trace over A , tr_A , is defined symmetrically.

11.2 The Real Composite Direction Disappears on Both Marginals

Proposition 11.1. $\text{tr}_B(J \otimes J) = 0$ and $\text{tr}_A(J \otimes J) = 0$.

Proof. Writing $J \otimes J$ in block form indexed by the first factor, each block is $J_{ij} \cdot J$ (the (i, j) entry of J times the full matrix J). $\text{tr}_B(J \otimes J)$ has (i, j) entry $J_{ij} \text{tr}(J) = 0$, since J is traceless. Hence $\text{tr}_B(J \otimes J) = 0$ identically. The argument for tr_A is the same with the roles of the two factors exchanged. $\square$

Corollary 11.2. For the family $\rho(c) = \frac{1}{4}(I \otimes I + c J \otimes J)$ of Example 10.2, both marginals are the maximally mixed single-rebit state for every $c \in [-1, 1]$:

$$\text{tr}_B \rho(c) = \text{tr}_A \rho(c) = \frac{1}{2}I, \quad (11.3)$$

independent of c .

Proof. $\text{tr}_B(\frac{1}{4}(I \otimes I + c J \otimes J)) = \frac{1}{4}(\text{tr}_B(I \otimes I) + c \text{tr}_B(J \otimes J)) = \frac{1}{4}(2I + 0) = \frac{1}{2}I$, using $\text{tr}_B(I \otimes I) = \text{tr}(I) \cdot I = 2I$ and Proposition 11.1. Identically for tr_A . $\square$

Remark 11.3. This is the payoff promised in Chapter 10. The entire one-real-parameter family of states $\rho(c)$, $c \in [-1, 1]$ — a whole interval of genuinely distinct, positivity-admissible global states — is completely invisible to an observer restricted to either marginal alone: every $\rho(c)$ looks identical (maximally mixed) from either subsystem's local perspective. The degree of freedom responsible for the failure of local tomography in Chapter 8 is exactly the degree of freedom annihilated by either partial trace:

globally physical $\not\Rightarrow$ locally recoverable.

(11.4)

Example 11.4 (The mechanism made explicit: Pauli decomposition). Writing $\rho(c)$ in the two-qubit Pauli tensor basis $\{I, X, Y, Z\} \otimes \{I, X, Y, Z\}$ (16 terms, coefficients $\text{tr}[(P \otimes Q)\rho(c)]/4$) makes the mechanism visible by inspection rather than by computation. Using $J = -i\sigma_y$ (Chapter 9's remark distinguishing the two), so that $J \otimes J = -\sigma_y \otimes \sigma_y$,

$$\rho(c) = \frac{1}{4}(I \otimes I - c \sigma_y \otimes \sigma_y), \quad (11.5)$$

with all fourteen remaining Pauli-tensor coefficients exactly zero — verified directly, not merely asserted. Since $\text{tr}(\sigma_y) = 0$ while $\text{tr}(I) = 2$, partial trace kills every term except the one built purely from identities: only $I \otimes I$ survives tr_A or tr_B , forcing both marginals to $\frac{1}{2}I$ regardless of c . The single surviving off-diagonal term, $\sigma_y \otimes \sigma_y$, is precisely the one Pauli-tensor combination invisible to any local Pauli measurement on either factor alone.

Ledger 11.5 (End of Chapter 11). Established: the partial trace is defined and shown to compute exactly the operator reproducing local measurement statistics; the $J \otimes J$ direction identified in Chapter 9 and bounded in Chapter 10 is shown to vanish under both partial traces, so the entire family $\rho(c)$ is marginally indistinguishable from the maximally mixed state despite being a genuine interval of distinct global states.

Not established: whether $\rho(c)$ (for $c \neq 0$) is entangled or merely classically correlated — that requires a definition of separability, not yet introduced, and is deferred to the next chapter. Nothing here yet distinguishes "locally invisible because of correlation" from "locally invisible because of some other structure"; that distinction is exactly what a separability criterion is for.

Chapter 12

Separability and Entanglement

Chapter 11 left one question open deliberately: is $\rho(c)$, for $c \neq 0$, entangled, or merely classically correlated in some other way invisible to marginals? This chapter answers it — and shows the answer depends on which scalar field is doing the composing, which is exactly the kind of question this book is positioned to ask.

12.1 Separability Relative to a Scalar Field

Definition 12.1 (Separability). Fix $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. A state $\rho \in \mathcal{D}_{d_A d_B}(\mathbb{F})$ is $\mathbb{F}$ -separable if it is a convex combination of $\mathbb{F}$ -product states,

$$\rho = \sum_k p_k \rho_A^{(k)} \otimes \rho_B^{(k)}, \quad p_k \geq 0, \quad \sum_k p_k = 1, \quad \rho_A^{(k)} \in \mathcal{D}_{d_A}(\mathbb{F}), \quad \rho_B^{(k)} \in \mathcal{D}_{d_B}(\mathbb{F}). \quad (12.1)$$

A state that is not $\mathbb{F}$ -separable is $\mathbb{F}$ -entangled.

The qualifier is essential: $\mathcal{D}_d(\mathbb{R}) \subsetneq \mathcal{D}_d(\mathbb{C})$ (Chapters 8–10), so the set of available local product states differs between the two theories, and separability is a property of a state relative to which local states are admitted — not an intrinsic property of a matrix alone.

12.2 The $J \otimes J$ Witness Over $\mathbb{R}$

Lemma 12.2. For any real symmetric $\rho_A \in \text{Herm}_{\mathbb{R}}(d)$, $\text{tr}(\rho_A J) = 0$ whenever $J^\top = -J$ is antisymmetric (same dimension).

Proof. $\text{tr}(\rho_A J) = \text{tr}((\rho_A J)^\top) = \text{tr}(J^\top \rho_A^\top) = \text{tr}(-J \rho_A) = -\text{tr}(J \rho_A) = -\text{tr}(\rho_A J)$ (using cyclicity of trace), so $\text{tr}(\rho_A J) = 0$. $\square$

Theorem 12.3 (Real entanglement witness). For every $\mathbb{R}$ -separable two-rebit state ρ , $\text{tr}[\rho(J \otimes J)] = 0$.

Proof. For a product state, $\text{tr}[(\rho_A \otimes \rho_B)(J \otimes J)] = \text{tr}(\rho_A J) \text{tr}(\rho_B J) = 0 \cdot 0 = 0$ by Lemma 12.2 (using the standard identity $\text{tr}[(A \otimes B)(C \otimes D)] = \text{tr}(AC) \text{tr}(BD)$). By linearity of trace, the same holds for any convex combination. $\square$

Proposition 12.4. $\text{tr}[\rho(c)(J \otimes J)] = c$.

Proof. $\text{tr}[\rho(c)(J \otimes J)] = \frac{1}{4}(\text{tr}(J \otimes J) + c \text{tr}[(J \otimes J)^2]) = \frac{1}{4}(0 + c \cdot 4) = c$, using $\text{tr}(J \otimes J) = 0$ and $(J \otimes J)^2 = I_4$ (Chapter 9), so $\text{tr}[(J \otimes J)^2] = \text{tr}(I_4) = 4$. $\square$

Corollary 12.5. $\rho(c)$ is $\mathbb{R}$ -separable if and only if $c = 0$.

Proof. If $c \neq 0$, Theorem 12.3 and Proposition 12.4 together rule out $\mathbb{R}$ -separability. If $c = 0$, $\rho(0) = \frac{1}{2}I \otimes \frac{1}{2}I$ is manifestly a real product state (indeed $\frac{1}{2}I \in \mathcal{D}_2(\mathbb{R})$), hence separable. $\square$

This is the four-chapter chain complete: $J \otimes J$ appears as an extra basis vector invisible to local product operators (Chapter 9), becomes a bounded physical coordinate $|c| \leq 1$ under positivity (Chapter 10), vanishes under both marginals (Chapter 11), and is now shown to be exactly an entanglement witness: the same coefficient invisible to every local measurement is precisely what makes $\rho(c)$, $c \neq 0$, entangled.

12.3 Complexification: The Same States, a Different Theory

Proposition 12.6. $J = -i\sigma_y$, so $J \otimes J = -\sigma_y \otimes \sigma_y$.

This is immediate from $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$. The point is that $J \otimes J$, unavailable as a product of real local observables (Chapter 9), is exactly -1 times a product of perfectly ordinary complex local observables, $\sigma_y \otimes \sigma_y$.

Theorem 12.7 (The same family is entirely $\mathbb{C}$ -separable). $\rho(c)$ is $\mathbb{C}$ -separable for every $c \in [-1, 1]$.

Proof. Let $|\pm y\rangle$ be the ± 1 -eigenvectors of σ_y . Since $J \otimes J = -\sigma_y \otimes \sigma_y$, and $\sigma_y \otimes \sigma_y$ has eigenvalue $+1$ on $\{|+y, +y\rangle, |-y, -y\rangle\}$ and eigenvalue -1 on $\{|+y, -y\rangle, |-y, +y\rangle\}$ (products of σ_y -eigenvalues), direct substitution gives the exact identity

$$\rho(c) = \frac{1+c}{4} |+y, -y\rangle\langle +y, -y| + \frac{1+c}{4} |-y, +y\rangle\langle -y, +y| + \frac{1-c}{4} |+y, +y\rangle\langle +y, +y| + \frac{1-c}{4} |-y, -y\rangle\langle -y, -y|. \quad (12.2)$$

Each term is a pure complex product state; the four weights are nonnegative and sum to 1 exactly when $|c| \leq 1$, which always holds (Chapter 10). This exhibits $\rho(c)$ as an explicit convex combination of $\mathbb{C}$ -product states for every admissible c . $\square$

Remark 12.8. This is stronger than a partial or asymptotic statement: the entire one-parameter family that is $\mathbb{R}$ -entangled for every $c \neq 0$ (Corollary 12.5) is $\mathbb{C}$ -separable across its whole range. Nothing about $\rho(c)$ itself changes — it is the same real 4×4 matrix throughout — only the theory doing the composing. Separability is not an intrinsic property of a state; it is a property of a state relative to the field over which local product states are admitted.

Scope. This is a precise claim about one family of matrices under Definition 12.1, not a general verdict that "entanglement isn't physically real." It does not say every entangled state in ordinary complex quantum mechanics becomes separable under some other field, and it does not say separability judgments are arbitrary or a matter of choice once a scalar field is fixed — within ordinary complex quantum mechanics, $\rho(c)$'s entanglement status (for whatever states arise there) is perfectly well-defined. What is field-relative is the comparison across candidate theories with different admissible local product states, which is the question this book's Parts I–II were built to ask.

Ledger 12.9 (End of Chapter 12). Established: $\rho(c)$ is $\mathbb{R}$ -entangled for every $c \neq 0$, exhibited by the explicit witness $J \otimes J$ (Corollary 12.5); the identical family is $\mathbb{C}$ -separable for every $c \in [-1, 1]$, exhibited by an explicit four-term product-state decomposition (Theorem 12.7); separability is therefore field-relative, not intrinsic to a state.

Not established: anything about entanglement in the full generality of $d > 2$ composite systems, multipartite entanglement, entanglement measures or their operational meaning, or the fate of quaternionic composite states under this classification (the overcompleteness of Chapter 9 suggests the very notion of a $\mathbb{H}$ -product state needs care before separability can even be defined there). These remain open.

Part III

Recovered Structure

Chapter 13

The Assumption Audit

This chapter derives nothing. Its purpose is to state plainly, in one place, exactly what Parts I and II have established and exactly what standard quantum mechanics contains that this book has not yet earned the right to use. After forty-some pages of successful reconstruction it becomes psychologically easy to feel that quantum mechanics has been derived. It has not — not yet — and this chapter exists so that no later chapter can quietly trade on that feeling. One clarification before the accounting itself: the ledger below records logical dependence, not historical dependence. An idea may have been known for decades — the Born rule predates every chapter of this book by a century — while still appearing late here, because in this reconstruction it depends on structure established only in the preceding chapters. The order of this book is not a claim about the order in which physics discovered these ideas.

13.1 What Has Been Established

Single-system foundations (Part I). The probability simplex and stochastic maps (Chapter 1); bistochastic matrices as a polytope generated by permutations, and bistochastic evolution as exactly majorization-compatible evolution with non-decreasing entropy (Chapter 2); the $n = 2$ degeneracy of orthostochastic and unistochastic matrices, and their strict separation at $n = 3$ via an explicit Hadamard-matrix obstruction (Chapter 3).

Compositional sufficiency (Part I, first axis). Probability-level composition discards information present in amplitude-level composition — but this alone does not select $\mathbb{C}$ over $\mathbb{R}$; a real lift shows the same obstruction (Chapter 4).

Global realizability (Part I, second axis). Exact frame closure around a cycle projects to no corresponding closure relation on observed transition probabilities (Chapter 5); edgewise real admissibility does not imply global real admissibility, proved via the triangle-gluing theorem and its mutually-unbiased corollary (Chapter 6), and generalized to arbitrary context graphs via spanning trees and cycle rank (§5.5). This is a genuine single-system obstruction to $\mathbb{R}^2$, not dependent on composite systems.

Extremal reformulation (Part II opening). The real/complex minimal witness at $d = 2$ is exactly K_3 , $m_\beta(\mathbb{R}, \mathbb{C}; 2) = 1$ (Chapter 7) — a restatement of the Part I result in extremal language, with several genuinely open extremal questions (higher- d witnesses, quaternionic separation, forbidden subgraphs, density thresholds) explicitly flagged as unresolved.

System composition (Part II, third axis). Local tomography, precisely defined via Hermitian-space dimension counting, holds exactly for $\mathbb{C}$ and fails in opposite directions for $\mathbb{R}$ and $\mathbb{H}$

(Chapter 8) — the classical Araki/Wootters/Hardy argument, not original to this book. For $\mathbb{C}$, tensor products of local operator bases literally span the composite Hermitian space; for $\mathbb{R}$, the missing dimension is identified concretely as $J \otimes J$ (Chapter 9). Positivity is derived operationally, and the $J \otimes J$ coefficient is shown to support a bounded physical family $|c| \leq 1$ (Chapter 10). Both partial traces annihilate this coefficient (Chapter 11). The same family is $\mathbb{R}$ -entangled for every $c \neq 0$ and $\mathbb{C}$ -separable throughout its entire range, via an explicit decomposition — separability is field-relative, not intrinsic to a state (Chapter 12).

13.2 What Remains

None of the above constitutes a derivation of quantum mechanics. The following pillars of the standard formalism have been used nowhere in Parts I–II and are not yet earned:

General complex Hilbert-space state structure. Everything proved about $\mathbb{C}$ so far concerns finite dimension, and the sharpest worked examples are $d = 2, 3$. The general claim that n -outcome quantum systems are described by rays or density operators on $\mathbb{C}^n$ has not been derived for general n , only motivated dimension-by-dimension.

The Born rule, in full. This book has been careful from Chapter 4 onward to keep four distinct statements separate and has proved none of the last three:

$$B_{ij} = |U_{ij}|^2 \quad (\text{proved, Ch. 3}), \quad p_i = |\psi_i|^2, \quad p(i|\psi) = |\langle i|\psi \rangle|^2, \quad p_i = \text{tr}(\rho P_i). \quad (13.1)$$

The bistochastic-matrix identity is not the same claim as the state-vector Born rule, which is not the same claim as the general projective-measurement rule, which is not the same claim as the density-operator/POVM rule.

Hermitian operators as the general observable class. Part II used $\text{Herm}_{\mathbb{C}}(d)$ as the ambient space for states, motivated by positivity of probabilities; it has not shown that observables in general (as opposed to states) must be Hermitian, nor derived the spectral decomposition $A = \sum_i a_i P_i$ and the eigenvalue rule for measurement outcomes.

Unitary dynamics and the Schrödinger equation. Nothing in this book has yet introduced continuous time evolution. The one-parameter group law $U(t+s) = U(t)U(s)$ and the generator theorem $U(t) = e^{-itH/\hbar} \Rightarrow i\hbar \partial_t |\psi\rangle = H|\psi\rangle$ remain to be derived, not assumed.

General measurement and POVM structure. Only rank-one projective tests have been used, to motivate positivity (Chapter 10). The general POVM formalism $p_i = \text{tr}(\rho E_i)$, $\sum_i E_i = I$, $E_i \geq 0$, has not been derived or even stated as a target.

13.3 Roadmap for Part III

Part III attacks these in dependency order. The Born rule is addressed first (Chapter 14), since the composite-system machinery of Part II already distinguishes exactly the four formulas above and Part III's task is to reconcile them rather than introduce new ones. Observables and spectral structure follow, motivated by the same admissibility logic used for states in Chapter 10. Continuous reversible composition is then used to derive the unitary group and Schrödinger's equation as a generator theorem, in the spirit of Chapter 5's treatment of discrete composition. POVMs are introduced last, as the generalization of projective measurement forced by considering measurements performed on a subsystem of a larger composite system — connecting directly back to the partial trace of Chapter 11.

Ledger 13.1 (Opening of Part III). Established: nothing new — this chapter is a compilation of Parts I–II, cross-referenced above, with no additional claims.

Not established: every item in §13.2. Part III exists to close these gaps one at a time, each under the same reconstruction-vs-recovery discipline used throughout: what is derived from what has already been proved, versus what is imported to check the result against the standard formalism of Sakurai–Napolitano [1] and Gross [3].

Chapter 14

The Born Hierarchy

Up to this point, this book has reconstructed constraints: what admissibility permits and forbids, without reference to the specific formulas a physics textbook opens with. From here onward, the project reverses direction — using those constraints to reconstruct the familiar formalism itself, formula by formula, with each one’s exact cost made visible rather than assumed. Chapter 13 flagged four distinct statements this book has been careful not to conflate. This chapter is not “the Born-rule derivation”; it is a dependency chain, with each link’s assumptions made explicit rather than absorbed silently into the next.

14.1 The General Context-Transition Identity

Chapters 5–6 established the context-transition interpretation of $B = |U|^2$ only for real two-dimensional frames (rotations in $\mathbb{R}^2$). The complex, general-dimension statement is an easy generalization, but it has not yet been written down, and Chapter 14 depends on it, so it is stated explicitly here rather than assumed.

Proposition 14.1 (Context-transition identity, general d). Let $F_A = (|a_1\rangle, \dots, |a_d\rangle)$ and $F_B = (|b_1\rangle, \dots, |b_d\rangle)$ be orthonormal bases of $\mathbb{C}^d$. Then $U := F_A^\dagger F_B$, with entries $U_{ij} = \langle a_i | b_j \rangle$, is unitary, and $B_{ij} := |\langle a_i | b_j \rangle|^2$ is unistochastic.

Proof. $(F_A^\dagger F_B)^\dagger (F_A^\dagger F_B) = F_B^\dagger F_A F_A^\dagger F_B = F_B^\dagger F_B = I$, using unitarity of F_A (so $F_A F_A^\dagger = I$) and of F_B ; hence U is unitary. $B_{ij} = |U_{ij}|^2$ is unistochastic by definition (Chapter 3). $\square$

14.2 From Contexts to Pure States

Proposition 14.1 is a statement about whole frames. It says nothing yet about a single normalized vector $|\psi\rangle$ carrying its own probabilities. Bridging this gap requires an explicit operational assumption, stated here for the first time in the book:

Assumption 14.2 (Pure states as frame-completable vectors). A prepared pure state $|\psi\rangle$ (normalized, $\langle\psi|\psi\rangle = 1$) can be operationally represented as the first vector of some orthonormal frame $F_\psi = (|\psi\rangle, |\psi_2\rangle, \dots, |\psi_d\rangle)$ — i.e. preparing $|\psi\rangle$ is representable as selecting one designated column of an admissible context.

This is a genuine addition to the book’s assumption base, not a consequence of Parts I–II: nothing proved so far distinguishes “a state” from “a context” as separate kinds of object, and this chapter’s first substantive move is to identify the former with a distinguished coordinate of the latter. Any vector can be completed to an orthonormal basis (Gram–Schmidt), so the mathematical content of Assumption 14.2 is only the operational claim that preparation is representable this way — worth keeping visible precisely because it is doing real work.

Theorem 14.3 (Pure-state Born rule). Given Assumption 14.2, and a measurement context $F_E = (|e_1\rangle, \dots, |e_d\rangle)$,

$$p_i := B_{i1} = |\langle e_i | \psi \rangle|^2, \quad (14.1)$$

where B_{i1} is the first column of the context-transition matrix between F_E and F_ψ .

Proof. Apply Proposition 14.1 with $A = E$, $B = \psi$: $U = F_E^\dagger F_\psi$ is unitary, with first column $U_{i1} = \langle e_i | \psi \rangle$ (the inner product of $|e_i\rangle$ with the first, designated column $|\psi\rangle$ of F_ψ). Then $B_{i1} = |U_{i1}|^2 = |\langle e_i | \psi \rangle|^2$ directly from the definition of the context-transition matrix. $\square$

Remark 14.4. This is the intended payoff: the pure-state Born rule is not an independent postulate about state vectors. It is literally one column of the already-established context-transition identity, once a state is identified with a frame via Assumption 14.2. The rest of F_ψ (the vectors $|\psi_2\rangle, \dots$) plays no further role and can be discarded; only the assumption that such a completion operationally exists was needed.

14.3 From Pure States to Density Operators

Part II already established convex mixtures of pure states as legitimate composite-system-independent objects (Chapter 10). For a classical mixture $\rho = \sum_a q_a |\psi_a\rangle\langle\psi_a|$ ($q_a \geq 0$, $\sum_a q_a = 1$), affine consistency under classical mixing — the probability of outcome i for a mixture is the corresponding mixture of probabilities — gives

$$p_i = \sum_a q_a |\langle e_i | \psi_a \rangle|^2. \quad (14.2)$$

Theorem 14.5 (Density-operator Born rule). $p_i = \text{tr}(\rho P_i)$, where $P_i = |e_i\rangle\langle e_i|$.

Proof. $|\langle e_i | \psi_a \rangle|^2 = \langle \psi_a | e_i \rangle \langle e_i | \psi_a \rangle = \text{tr}(|\psi_a\rangle\langle\psi_a| P_i)$ (cyclicity of trace). Summing with weights q_a and using linearity of trace, $p_i = \text{tr}\left(\left(\sum_a q_a |\psi_a\rangle\langle\psi_a|\right) P_i\right) = \text{tr}(\rho P_i)$. $\square$

Remark 14.6. No new assumption was needed for this step beyond Theorem 14.3 and the affine-mixture structure Part II already earned. This is the cleanest link in the chain.

14.4 What Is Not Yet Earned: POVMs

$$p_i = \text{tr}(\rho E_i), \quad E_i \succeq 0, \quad \sum_i E_i = I \quad (14.3)$$

is a strictly more general statement than Theorem 14.5: it permits E_i that are not rank-one projectors, and in particular need not come from any single orthonormal context at all. Nothing proved in this chapter licenses that generality. The standard route — representing a generalized measurement as an ordinary projective measurement on a larger, composite system (a dilation) and invoking the partial trace of Chapter 11 — requires its own argument and is deferred rather than assumed here.

Ledger 14.7 (End of Chapter 14). Established: a fully explicit dependency chain,

$$\boxed{|U_{ij}|^2 \text{ (Ch.3)} \longrightarrow |\langle e_i | \psi \rangle|^2 \text{ (Thm.14.3)} \longrightarrow \text{tr}(\rho P_i) \text{ (Thm.14.5)}} \quad (14.4)$$

with each arrow's cost stated: the first arrow costs exactly Assumption 14.2 (pure states as frame-completable vectors, a genuinely new addition to the book's assumption base); the second arrow costs nothing beyond what Part II already proved.

Not established: the POVM generalization $p_i = \text{tr}(\rho E_i)$, deferred to a later chapter pending a

dilation argument; the general complex Hilbert-space state structure for arbitrary n (this chapter's propositions are dimension-generic but the book's worked examples remain concentrated at $d = 2, 3$); Hermitian observables as the general observable class, and spectral decomposition, remain untouched by this chapter and are next.

Chapter 15

Observables and Spectral Decomposition

Chapter 14 established probabilities for a fixed orthonormal context, $p_i = \text{tr}(\rho P_i)$ with $P_i = |e_i\rangle\langle e_i|$. This chapter asks what an observable is, and why Hermitian operators are the right general class — not by declaration, but via an explicit operational bridge. The route runs in two steps: first widen the rank-one projectors of Chapter 14 to allow coarser, physically realistic measurements with grouped outcomes, then ask what it means to attach a numerical value to each outcome of such a measurement. That second step is where Hermitian operators enter — as a consequence of the attachment, not as a new ingredient assumed alongside it.

15.1 Sharp Measurements and Coarse-Grained Outcomes

Definition 15.1 (Sharp measurement). A sharp measurement is a family of mutually orthogonal projectors $\{P_a\}$ with $\sum_a P_a = I$ ($P_a P_b = \delta_{ab} P_a$), representing mutually exclusive, jointly exhaustive outcomes.

Chapter 14 proved $p_i = \text{tr}(\rho P_i)$ only for rank-one P_i (a full orthonormal basis). The extension to Definition 15.1's coarser projectors costs nothing new:

Proposition 15.2 (Coarse-grained Born rule). For a sharp measurement $\{P_a\}$ where each P_a is a sum of rank-one basis projectors, $P_a = \sum_{i \in S_a} P_i$ for disjoint index sets S_a partitioning $\{1, \dots, d\}$, $\text{tr}(\rho P_a) = \sum_{i \in S_a} p_i$.

Proof. Immediate from linearity of trace and Theorem 14.5. □

15.2 The Operational Bridge

Definition 15.3 (Observable). Given a sharp measurement $\{P_a\}$ and a real value a assigned to each outcome, the corresponding observable is

$$A = \sum_a a P_a. \tag{15.1}$$

Since each P_a is Hermitian and $a \in \mathbb{R}$, A is Hermitian. This is one direction of the bridge: packaging a real-labeled sharp measurement produces a Hermitian operator, by construction, not by postulate.

Theorem 15.4 (Spectral theorem, cited). Every Hermitian operator A on a finite-dimensional complex Hilbert space admits a decomposition $A = \sum_a a P_a$ with real a and mutually orthogonal projectors P_a summing to I .

This is a classical result of linear algebra, entirely independent of this book’s reconstruction, and is not re-derived here; it is cited exactly as Chapters 2 and 8 cited Birkhoff–von Neumann’s antecedents and Araki’s parameter-counting argument. What Theorem 15.4 supplies is the converse of Definition 15.3: every Hermitian A unpacks into exactly the kind of real-labeled sharp measurement the definition packages.

Remark 15.5. The bridge is therefore an equivalence, not a declaration: real-valued sharp projective measurements and Hermitian operators are the same data, related by Definition 15.3 in one direction and Theorem 15.4 in the other. ”Observables are Hermitian” is the statement of this equivalence, not an independent postulate layered on top of the state-space structure of Parts I–II. But the equivalence has a hidden cost worth naming rather than absorbing silently: it holds only because Definition 15.3 already assumed outcomes are labeled by real numbers. Nothing established so far forces this — it is a modeling choice (numerical outcomes are what get recorded, compared, and averaged), consistent with the book’s discipline of not letting assumptions go unmarked. Complex or otherwise non-real labelings are mathematically available and simply produce non-Hermitian (but still well-defined) operators; the restriction to real labels is what makes A Hermitian, not a fact independently established about the physical world.

15.3 The Eigenvalue Rule as a Corollary

Proposition 15.6 (Eigenvalue rule). If $A|\psi\rangle = a|\psi\rangle$ for normalized $|\psi\rangle$, then $\Pr(a \mid \psi) = 1$.

Proof. $A|\psi\rangle = a|\psi\rangle$ places $|\psi\rangle$ in the a -eigenspace, which is exactly the range of the spectral projector P_a (Theorem 15.4); since $|\psi\rangle$ is normalized and lies entirely in that eigenspace, $P_a|\psi\rangle = |\psi\rangle$. By Theorem 14.3 (equivalently Proposition 15.2 applied to a pure state), $\Pr(a \mid \psi) = \langle\psi|P_a|\psi\rangle = \langle\psi|\psi\rangle = 1$. $\square$

No new probability postulate was needed to get here. ”An eigenstate returns its eigenvalue with certainty” — a rule ordinarily stated as its own axiom — turns out to be a corollary of Chapter 14’s already-proved Born rule, once Definition 15.3 pins down what ”the eigenvalue” means operationally: nothing more than which spectral projector a given state happens to lie inside.

Ledger 15.7 (End of Chapter 15). Established: an operational equivalence between real-valued sharp projective measurements and Hermitian operators (Definition 15.3, Theorem 15.4); the eigenvalue rule as a corollary of Chapter 14’s Born rule, not an independent postulate (Proposition 15.6).

Not established: why outcome labels must be real (flagged explicitly as an assumption, not derived, Remark 15.5); the spectral theorem itself is cited from classical linear algebra, not proved within this book’s reconstruction; compatible observables, simultaneous diagonalization, and commutators are untouched; the status of general (non-Hermitian, non-normal) operators is not addressed. Continuous dynamics — how observables and states evolve in time — is the subject of the next chapter.

Chapter 16

Unitary Dynamics and Schrödinger's Equation

16.1 An Explicit New Assumption: Continuous Reversible Evolution

Parts I–II established that coherent, reversible transformations between discrete contexts are represented by unitary lifts (Chapters 4–6). Nothing proved so far concerns physical time. Connecting the two requires a genuinely new assumption:

Assumption 16.1 (Continuous reversible time evolution). A closed system's evolution over time is represented by a strongly continuous one-parameter family of unitary operators $U(t)$ satisfying

$$U(t+s) = U(t)U(s), \quad U(0) = I. \quad (16.1)$$

This is not derived from the discrete admissibility framework of Parts I–II — it imports continuity in time as new physical content, using "unitary" only because Parts I–II already established unitary operators as the coherent lifts of admissible transformations. The group law itself is the same composition law studied throughout the book (Chapter 5's exact closure, Chapter 4's composition rule), now imposed on a continuous parameter rather than a discrete sequence of contexts.

16.2 The Generator Theorem

In finite dimension, Assumption 16.1 has an elementary consequence — no need for the general (infinite-dimensional) machinery of Stone's theorem.

Theorem 16.2 (Finite-dimensional generator theorem). Given Assumption 16.1 with $U(t)$ differentiable at $t = 0$, there is a Hermitian operator H and real constant $\hbar$ such that

$$U(t) = e^{-itH/\hbar}. \quad (16.2)$$

Proof. Let $K := \frac{d}{dt}U(t)|_{t=0}$. Differentiating $U(t+s) = U(t)U(s)$ with respect to s at $s = 0$ gives $\frac{d}{dt}U(t) = U(t)K$, a linear ODE with solution $U(t) = e^{Kt}$ (using $U(0) = I$). Differentiating $U(t)^\dagger U(t) = I$ at $t = 0$ gives $K^\dagger + K = 0$, i.e. K is anti-Hermitian, so $K = -iH/\hbar$ for some Hermitian H and real scale $\hbar$ (any decomposition of an anti-Hermitian operator as $-i$ times a Hermitian one, with the scale factor absorbed into H or left explicit as $\hbar$). $\square$

Remark 16.3. This is genuinely elementary — a finite-dimensional linear ODE argument, not an import of Stone's theorem (which handles the harder infinite-dimensional case this book's finite- d scope does not need). The differentiability assumed at $t = 0$ is itself part of Assumption 16.1's content; strong continuity alone, without differentiability, requires the fuller argument this book does not attempt.

16.3 Schrödinger's Equation

Corollary 16.4 (Schrödinger's equation). Writing $|\psi(t)\rangle := U(t)|\psi(0)\rangle$,

$$i\hbar \frac{d}{dt}|\psi(t)\rangle = H|\psi(t)\rangle. \quad (16.3)$$

Proof. $\frac{d}{dt}|\psi(t)\rangle = \frac{d}{dt}U(t)|\psi(0)\rangle = -\frac{i}{\hbar}H U(t)|\psi(0)\rangle = -\frac{i}{\hbar}H|\psi(t)\rangle$, using Theorem 16.2 and $\frac{d}{dt}e^{-itH/\hbar} = -\frac{i}{\hbar}H e^{-itH/\hbar}$ (differentiating the matrix exponential). Multiplying by $i\hbar$ gives the stated equation. $\square$

Remark 16.5. Schrödinger's equation is, on this route, the infinitesimal form of the composition law of Assumption 16.1 — not an independent postulate stacked on top of the state-space structure recovered in Chapters 14–15.

16.4 What H Is Not (Yet)

Theorem 16.2 produces a Hermitian generator and a real scale $\hbar$. It does not say that H is energy, nor that $\hbar$ is Planck's constant. Those identifications are additional physical input — typically made via correspondence with classical Hamiltonian mechanics (the generator of time translation is identified with energy through Noether's theorem or the classical Hamiltonian's role as generator of time evolution in the classical limit) — and are not established by anything in this chapter. The generator theorem gives the form $i\hbar d_t|\psi\rangle = H|\psi\rangle$; it does not by itself explain why the object called H deserves to be called energy.

Ledger 16.6 (End of Chapter 16). Established: given Assumption 16.1 (stated explicitly as new, not derived), the finite-dimensional generator theorem (Theorem 16.2) and Schrödinger's equation as its infinitesimal form (Corollary 16.4), via an elementary matrix-ODE argument rather than the general Stone's theorem.

Not established: that physical time evolution actually satisfies Assumption 16.1 — this is imported, not derived from Parts I–II's discrete admissibility framework; that the generator H is energy or that $\hbar$ is Planck's constant, both of which require additional physical identification not attempted here; anything about open-system or non-unitary dynamics, which would require the channel/Kraus formalism this book has not developed.

Chapter 17

Generalized Measurement and Dilation

Chapter 14 deliberately deferred one statement: $p_i = \text{tr}(\rho E_i)$ for general effects E_i , not necessarily rank-one projectors. This chapter closes that gap, and shows it reconnects directly to Chapter 11's partial trace — the two constructions are complementary directions through the same system-plus-environment architecture.

17.1 Effects and POVMs

Definition 17.1 (POVM). A positive operator-valued measure (POVM) is a family of operators $\{E_i\}$ with $0 \preceq E_i \preceq I$ and $\sum_i E_i = I$. The rule

$$p_i = \text{tr}(\rho E_i) \quad (17.1)$$

generalizes Chapter 15's sharp measurements: a sharp measurement is the special case $E_i = P_i$ with $P_i^2 = P_i$; a general E_i need not be a projector, and need not be diagonal in any single orthonormal basis shared by all the other E_j .

17.2 Dilation: Every POVM Is a Sharp Measurement, Elsewhere

Theorem 17.2 (Naimark dilation, finite-dimensional). For any POVM $\{E_i\}_{i=1}^n$ on $\mathcal{H}$, there is an ancilla Hilbert space $\mathbb{C}^n$, a reference state $|a_0\rangle \in \mathbb{C}^n$, a unitary U on $\mathcal{H} \otimes \mathbb{C}^n$, and a sharp measurement $\{\Pi_i = I_{\mathcal{H}} \otimes |i\rangle\langle i|\}$ on $\mathcal{H} \otimes \mathbb{C}^n$, such that for every state ρ on $\mathcal{H}$,

$$\text{tr}(\rho E_i) = \text{tr} \left[U(\rho \otimes |a_0\rangle\langle a_0|) U^\dagger \Pi_i \right]. \quad (17.2)$$

Proof. Define $V : \mathcal{H} \rightarrow \mathcal{H} \otimes \mathbb{C}^n$ by

$$V|\psi\rangle = \sum_{i=1}^n (\sqrt{E_i}|\psi\rangle) \otimes |i\rangle, \quad (17.3)$$

using the unique positive square root of each $E_i \succeq 0$. For any $|\phi\rangle, |\psi\rangle$,

$$\langle V\phi | V\psi \rangle = \sum_i \langle \phi | \sqrt{E_i} \sqrt{E_i} | \psi \rangle = \langle \phi | \left(\sum_i E_i \right) | \psi \rangle = \langle \phi | \psi \rangle, \quad (17.4)$$

using $\sum_i E_i = I$: V is an isometry. Fix $|a_0\rangle := |1\rangle \in \mathbb{C}^n$ (any basis vector). Since V is an isometry, the assignment $U(|\psi\rangle \otimes |a_0\rangle) := V|\psi\rangle$ is well-defined and inner-product-preserving on the subspace $\mathcal{H} \otimes |a_0\rangle$; by the standard extension theorem for isometries between finite-dimensional Hilbert spaces (extend an orthonormal basis of the range to a full orthonormal basis of the codomain), U extends to a unitary on the whole space $\mathcal{H} \otimes \mathbb{C}^n$.

For a pure state $|\psi\rangle$, a direct computation (expanding $(I \otimes |i\rangle\langle i|)V|\psi\rangle = (\sqrt{E_i}|\psi\rangle) \otimes |i\rangle$ and taking the inner product with $V|\psi\rangle$) gives

$$\langle V\psi|\Pi_i|V\psi\rangle = \langle\psi|\sqrt{E_i}\sqrt{E_i}|\psi\rangle = \langle\psi|E_i|\psi\rangle = \text{tr}(|\psi\rangle\langle\psi|E_i), \quad (17.5)$$

and since $U(|\psi\rangle \otimes |a_0\rangle) = V|\psi\rangle$, this is exactly $\text{tr}[U(|\psi\rangle\langle\psi| \otimes |a_0\rangle\langle a_0|)U^\dagger \Pi_i]$. Both sides of the claimed identity are linear in ρ , so the pure-state case extends to general ρ by writing $\rho = \sum_k q_k |\psi_k\rangle\langle\psi_k|$ and summing. $\square$

Remark 17.3. This makes precise the sense in which every generalized measurement is an ordinary sharp measurement performed elsewhere: couple the system to an ancilla prepared in a fixed reference state, evolve unitarily, and measure projectively on the composite. Nothing about the POVM formalism is a new kind of physical process; it is the statistics of Chapters 14–15, viewed from a subsystem of a larger sharp measurement.

17.3 The Same Architecture as Partial Trace

Remark 17.4 (Two directions through one construction). Chapter 11 asked: given a joint state on $\mathcal{H} \otimes \mathcal{H}'$, what does a local observer restricted to $\mathcal{H}$ see? The partial trace answers this by discarding $\mathcal{H}'$ entirely. This chapter asks the complementary question: given only local access to $\mathcal{H}$, what joint constructions on $\mathcal{H} \otimes \mathcal{H}'$ could have produced a given set of local statistics? Theorem 17.2 answers this by exhibiting $\mathcal{H}' = \mathbb{C}^n$ explicitly. The two constructions are literally inverse in spirit — one forgets an already-composite system’s inaccessible part; the other manufactures the composite that a given (possibly non-projective) local measurement scheme could have come from — and both rest on exactly the same system-plus-ancilla architecture introduced for entanglement in Chapter 12.

Ledger 17.5 (End of Chapter 17). Established: every finite POVM admits an explicit Naimark dilation into a sharp measurement on a system-plus-ancilla composite (Theorem 17.2, proved constructively, not merely cited); this closes the last gap flagged in Chapter 14’s ledger.

Not established: anything about POVMs with a continuum of outcomes (this book’s finite-dimensional scope stops at finite n); the operational question of which unitary U and which ancilla a given real-world apparatus actually realizes is not addressed — Theorem 17.2 guarantees some dilation exists, not that it is unique or physically distinguished; instruments and post-measurement state updates (the analogue of $\rho \mapsto \sqrt{E_i}\rho\sqrt{E_i}^\dagger / \text{tr}(\rho E_i)$) are not derived here.

Chapter 18

Final Audit: What This Book Actually Established

This chapter derives nothing. Chapter 13 took a snapshot after Part II; this chapter classifies the entire book, statement by statement, into four categories:

- [D] Derived — proved within this book’s reconstruction, from admissibility constraints already established, with no new physical assumption.
- [R] Equivalent reformulation — restates an earlier result in different language; adds no independent content.
- [A] Independently assumed — a genuine physical or operational principle introduced because the admissibility logic alone did not force it, flagged explicitly at the point of introduction.
- [U] Unresolved — an open question this book raises but does not answer.

The purpose is falsifiability about the book’s own achievement: a reader should be able to check every claim’s category against where it was proved, assumed, or left open, rather than take “quantum mechanics has been reconstructed” on faith.

18.1 Single-System Foundations (Part I)

- [D] Probability simplex, stochastic maps, deterministic maps as extreme points (Ch. 1).
- [D] Birkhoff–von Neumann theorem; Hardy–Littlewood–Pólya majorization; entropy non-decrease as a structural corollary (Ch. 2).
- [D] $n = 2$ orthostochastic/unistochastic degeneracy; $n = 3$ incompleteness via the Hadamard-matrix obstruction (Ch. 3).
- [D] Probability-level composition is not a homomorphism of π (Ch. 4).
- [U] Whether interference alone selects $\mathbb{C}$ over $\mathbb{R}$ — explicitly answered no within Ch. 4 itself; resolved (as a negative result) rather than left open.
- [D] Exact frame closure around a cycle; the general graph realizability theorem (spanning tree + fundamental-cycle checks) (Ch. 5).
- [D] Triangle-gluing theorem and the real-MUB corollary (Ch. 6) — the single-system obstruction to $\mathbb{R}^2$, independent of composite systems.
- [U] Whether every non-orthostochastic unistochastic matrix fails to be orthostochastic for the same structural reason as the $n = 3$ DFT example, or whether distinct mechanisms are at play in general (Ch. 3).

18.2 System Composition (Part II)

- [R] Minimal-witness reformulation, $m_\beta(\mathbb{R}, \mathbb{C}; 2) = 1$ at K_3 (Ch. 7) — restates Ch. 6 in extremal language.
- [U] Higher- d minimal witnesses; any quaternionic minimal witness; forbidden-subgraph characterizations; density thresholds (Ch. 7, §7.3) — an explicit research programme, not results.
- [A] Local tomography, as a principle imposed and tested rather than derived from Part I (Ch. 8).
- [D] The dimension identity distinguishing $\mathbb{R}, \mathbb{C}, \mathbb{H}$ under local tomography, given that principle (Ch. 8).
- [D] Tensor products of local bases span the composite space for $\mathbb{C}$; the concrete missing $\mathbb{R}$ direction $J \otimes J$ (Ch. 9).
- [D] Positivity derived operationally from nonnegative test probabilities; the $|c| \leq 1$ bound (Ch. 10).
- [D] Partial trace defined and shown to reproduce local statistics; $J \otimes J$ invisible to both marginals (Ch. 11).
- [D] Field-relative separability: the $J \otimes J$ witness for $\mathbb{R}$; the explicit $\mathbb{C}$ -separable decomposition for the entire range (Ch. 12).
- [U] Multipartite entanglement, entanglement measures, $d > 2$ composite systems in general; what "separable" even means for $\mathbb{H}$, given Ch. 9's overcompleteness (Ch. 12).

18.3 Recovered Formalism (Part III)

- [D] General- d context-transition identity (Prop. 14.1) — a new, easy generalization of the real 2D case, not previously written down explicitly.
- [A] Pure states as frame-completable vectors (Assumption 14.2, Ch. 14) — a genuine new operational assumption identifying "a state" with a distinguished frame coordinate.
- [D] Pure-state Born rule, given that assumption; density-operator Born rule, at no further cost beyond Part II's mixture structure (Ch. 14).
- [D] The Hermitian/real-sharp-measurement equivalence (one direction by construction, the other by the cited spectral theorem); the eigenvalue rule as a corollary (Ch. 15).
- [A] Real-valued outcome labels (Remark 15.5) — flagged explicitly as a modeling choice, not derived.
- [U] The spectral theorem itself — cited from classical linear algebra, not proved within this book's reconstruction (Ch. 15). Compatible observables and commutators.
- [A] Continuous reversible time evolution as a strongly continuous unitary group (Assumption 16.1, Ch. 16) — imported, not derived from Parts I–II's discrete framework.
- [D] The finite-dimensional generator theorem and Schrödinger's equation as its derivative, given that assumption (Ch. 16).

- [U] That the generator H is energy, or that $\hbar$ is Planck’s constant — flagged explicitly as requiring additional physical identification not attempted (Ch. 16). Open-system/non-unitary dynamics and the Kraus formalism.
- [D] Naimark dilation for finite POVMs, proved constructively (Ch. 17).
- [U] Continuum-outcome POVMs; uniqueness or physical distinguishedness of a given dilation; instruments and post-measurement state updates (Ch. 17).

18.4 A Note on Recent Literature: Representation Versus Structure

Two papers appearing during the writing of this book bear directly on the Kronecker-product convention used throughout Chapters 8 and 9, and are worth citing precisely rather than leaving this book silent about.

Hoffreumon and Woods [10] construct a real-number quantum theory reproducing every prediction of standard complex quantum theory, responding to the 2021 falsification argument of Renou et al. [9]. Their construction replaces the Kronecker product with an alternative representation $\otimes_r$ of the abstract tensor product, satisfying $\Gamma(H \otimes L) = \Gamma(H) \otimes_r \Gamma(L)$ for a realification map Γ decomposing a Hermitian matrix $H = H^{\text{Sy}} + iH^{\text{An}}$ into real symmetric and antisymmetric parts. The real matrix standing in for i in their construction is

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad (18.1)$$

identical to the J of Chapters 9–12. Barrios Hita, Trushechkin, Kampermann, Epping, and Bruß [11] reach the same empirical conclusion via a different construction, quotienting the ordinary real tensor product by the phase-redistribution redundancy $i|\psi_A\rangle \otimes (-i)|\psi_B\rangle = |\psi_A\rangle \otimes |\psi_B\rangle$ carries in the complex theory.

Remark 18.1 (What changes, and what does not). Both papers confirm that the Kronecker product is one admissible matrix representation of composition, selected by convention rather than forced by locality or linearity alone — precisely the distinction this book’s admissibility methodology asks about everywhere else, and had not asked about here. Chapters 8 and 9 used the Kronecker product throughout without flagging it as a choice; every claim proved there remains correct for that representation, but the audit should not be read as having shown Kronecker composition to be the unique or necessary choice.

This point is older than either paper, and worth stating precisely rather than leaving implicit. That a complex number is equivalent to a real 2×2 rotation-scaling matrix, $a + bi \leftrightarrow \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$ — Needham’s amplitwist: multiplication by $z = re^{i\theta}$ as simultaneous scaling by r and rotation by θ [12] — has nothing to do with quantum mechanics and predates these papers by decades. What is genuinely new in Hoffreumon–Woods and Barrios Hita et al. is not this realification (classical), but the demonstration that a full multipartite composition law can be built on top of it while preserving locality and reproducing every quantum prediction — a nontrivial claim about composition, not merely a restatement of the classical fact that i is a rotation-scaling operator in disguise. The two should not be conflated: the elementary observation licenses no conclusion about quantum composition on its own, and the papers’ technical content is precisely the part the elementary observation does not supply.

What does not change is the assumption count of §18.5. Local tomography is not derived by either paper from weaker principles; Hoffreumon and Woods obtain it for their real theory via a one-to-one correspondence with complex quantum theory’s own local tomography, which is a transfer argument, not a derivation from representation locality alone. The distinct question these papers answer — whether complex scalars are necessary, given that some coherent structure equivalent to $J^2 = -I$ survives into the composition law regardless — is a different,

and in a precise sense narrower, claim than whether the admissibility constraints of Parts I–II force **C**. This book’s real-MUB and local-tomography arguments (Chapters 6, 8) are unaffected by either paper: they rule out $\mathbb{R}$ under the Kronecker representation specifically, and whether they still rule out $\mathbb{R}$ under a $\otimes_r$ -style representation is a genuinely open question this book does not address.

Ledger 18.2 (Addendum). Not established: whether Chapters 8 and 9’s results survive replacement of the Kronecker product with a non-Kronecker representation of composition in the style of Hoffreumon–Woods or Barrios Hita et al. This is left as an explicit open question rather than assumed away, added to the record of §7.3 and §13.2.

18.5 The Assumptions, Collected

Stripped of derivation and restated together, this is the complete list of what this book’s reconstruction has added to bare admissibility reasoning, beyond what Parts I–II derive from constraint alone:

1. Local tomography (Ch. 8) — tested, not forced.
2. Pure states are frame-completable vectors (Ch. 14).
3. Measurement outcomes are labeled by real numbers (Ch. 15).
4. Time evolution is a strongly continuous unitary group (Ch. 16).

Four assumptions. Everything else in the recovered formalism — the Born rule in all its forms, the Hermitian-observable correspondence, the eigenvalue rule, Schrödinger’s equation, and the Naimark dilation of POVMs — is derived from these four plus Parts I–II’s admissibility structure, not independently postulated. Whether four is a small number is a matter of comparison the reader can make directly against Sakurai–Napolitano’s [1] or Gross’s [3] postulate lists, cited throughout this book precisely so that comparison is possible.

Scope. A small number of assumptions is not the same claim as zero assumptions, and it is not the same claim as inevitability. Four independently chosen principles, none derived from anything more primitive, is compatible with several different theories of quantum mechanics being equally well-founded — it is not a proof that this particular four is the unique starting point physics could have had. Nothing in this count should be read as “quantum mechanics is the only way the pieces fit together”; it should be read as “this is exactly what standard quantum mechanics costs, on this specific route, stated precisely enough to compare against other routes” — including the routes discussed in §18.4 below, which reach empirically equivalent theories via a different, likewise small, assumption set.

Ledger 18.3 (Closing entry). This chapter adds no new mathematics, with one exception: §18.4 notes that the Kronecker-product convention used in Chapters 8–9 is a representation choice, not a forced one, per recent literature — itself an audit-relevant fact, not a new derivation. Otherwise: every derived statement traces to a cited line, every assumption is numbered above, and every unresolved question is listed rather than quietly dropped. Any future chapter that treats an [A]-tagged or [U]-tagged item as though it were [D] should be read as an error against this audit, not a new result.

Chapter 19

The Amplitwist Foundations of Complex Structure

Chapter 18's addendum (§18.4) cited recent literature showing that the Kronecker-product convention used from Chapter 8 onward is a representation choice, and flagged the classical fact that complex multiplication is itself a rotation in disguise. This chapter makes that fact fully explicit and self-contained — proved from elementary linear algebra, not imported by citation — and extends it to the composite-system construction underlying Barrios Hita et al.'s real-valued quantum theory [11].

19.1 Scaling and Rotation Combined

Chapter 5 used the rotation matrix $R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ extensively. A scaling by a factor $s > 0$ is $S(s) = sI$. The two commute (both are polynomials in a single 2×2 matrix structure), so it is natural to ask whether they combine into a single operator.

Definition 19.1 (Amplitwist). An amplitwist is $A = sR(\theta)$ for $s > 0$, $\theta \in \mathbb{R}$. Needham coined the term for exactly this combined operation — simultaneous amplitude scaling and angular twist [12].

Theorem 19.2 (Amplitwist composition). If $A_1 = s_1R(\theta_1)$ and $A_2 = s_2R(\theta_2)$, then $A_1A_2 = s_1s_2R(\theta_1 + \theta_2)$: amplitudes multiply, angles add.

Proof. $A_1A_2 = s_1R(\theta_1)s_2R(\theta_2) = s_1s_2R(\theta_1)R(\theta_2) = s_1s_2R(\theta_1 + \theta_2)$, using that scalars commute with matrix multiplication and the rotation composition law $R(\theta_1)R(\theta_2) = R(\theta_1 + \theta_2)$ (immediate from the angle-addition formulas for sine and cosine). $\square$

Proposition 19.3 (Amplitwists form a group). The set of amplitwists is closed under composition, contains an identity $I = 1 \cdot R(0)$, and every amplitwist $A = sR(\theta)$ has inverse $A^{-1} = \frac{1}{s}R(-\theta)$.

Proof. Closure is Theorem 19.2. $I \cdot sR(\theta) = sR(\theta)$ since $R(0) = I_2$. For the inverse, $(sR(\theta))(\frac{1}{s}R(-\theta)) = R(\theta)R(-\theta) = R(0) = I$ by Theorem 19.2. $\square$

19.2 Amplitwists Are Complex Multiplication

Identify $\mathbb{R}^2$ with $\mathbb{C}$ via $(x, y) \leftrightarrow x + iy$. The action of an amplitwist on this identification is exactly multiplication by a complex number.

Proposition 19.4 (Amplitwists realize complex multiplication). Under $(x, y) \leftrightarrow x + iy$, the amplitwist $A = sR(\theta)$ acts as multiplication by $z = se^{i\theta}$: $A\begin{pmatrix} x \\ y \end{pmatrix} \leftrightarrow z(x + iy)$.

Proof. $A\begin{pmatrix} x \\ y \end{pmatrix} = s\begin{pmatrix} x \cos \theta - y \sin \theta \\ x \sin \theta + y \cos \theta \end{pmatrix}$, corresponding to $s[(x \cos \theta - y \sin \theta) + i(x \sin \theta + y \cos \theta)] = s(\cos \theta + i \sin \theta)(x + iy) = se^{i\theta}(x + iy)$ by direct expansion and Euler's formula. $\square$

Every nonzero complex number is uniquely an amplitwist: $z = x + iy \neq 0$ gives $s = \sqrt{x^2 + y^2} > 0$ and $\theta = \text{atan2}(y, x)$, so Theorem 19.2's composition law, translated through Proposition 19.4, is the complex multiplication rule $z_1 z_2 = s_1 s_2 e^{i(\theta_1 + \theta_2)}$ — not analogous to it, but the same statement in different notation. Complex multiplication was never a separate fact requiring its own justification; it is amplitwist composition, proved above using nothing but real 2×2 matrices.

19.3 The Generator J

Proposition 19.5 (The quarter-turn generator). $J := R(\pi/2) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ satisfies $J^2 = -I$, and $aI + bJ$ for $a, b \in \mathbb{R}$ reproduces complex arithmetic exactly: $(aI + bJ)(cI + dJ) = (ac - bd)I + (ad + bc)J$, matching $(a + ib)(c + id) = (ac - bd) + i(ad + bc)$.

Proof. $J^2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I$. Expanding $(aI + bJ)(cI + dJ) = acI + adJ + bcJ + bdJ^2 = (ac - bd)I + (ad + bc)J$ using $J^2 = -I$ and commutativity of I, J (both are polynomials in J). $\square$

This J is exactly the antisymmetric generator of Chapters 9–12: the direction invisible to real local observables (Chapter 9), the bounded degree of freedom of Chapter 10's $\rho(c)$, and the entanglement witness of Chapter 12. It is also, identically, the real matrix Hoffreumon and Woods use to represent i [10] (Chapter 18, §18.4). One object recurs across four chapters and two independent lines of external literature: not a coincidence needing explanation, but confirmation that J is the right invariant to have been tracking all along.

Scope. "The right invariant" is a claim about J , not a claim that complex scalars are the only way to write it. The whole point of this chapter is that J is expressible equally well as a literal 2×2 real matrix — the convergence above is evidence that this book's earlier chapters were tracking a real structural fact, not evidence that the fact must be written using complex numbers specifically.

19.4 Composite Systems: A Self-Derived Illustration

Single-system realification is elementary; the difficulty is composition, and it is worth deriving the obstruction directly rather than citing it.

For $|\psi\rangle \in \mathbb{C}^d$, define the flag map $\mathcal{S}(|\psi\rangle) := \text{Re } |\psi\rangle \otimes |0\rangle_F + \text{Im } |\psi\rangle \otimes |1\rangle_F \in \mathbb{R}^d \otimes \mathbb{R}^2$. A global phase becomes a flag rotation:

Proposition 19.6. $\mathcal{S}(e^{i\alpha}|\psi\rangle) = (I_d \otimes R_F(\alpha))\mathcal{S}(|\psi\rangle)$, where $R_F(\alpha)$ acts on the two-dimensional flag factor.

Proof. $\text{Re}(e^{i\alpha}\psi) = \cos \alpha \text{Re } \psi - \sin \alpha \text{Im } \psi$ and $\text{Im}(e^{i\alpha}\psi) = \sin \alpha \text{Re } \psi + \cos \alpha \text{Im } \psi$ (expand $e^{i\alpha} = \cos \alpha + i \sin \alpha$ and collect real/imaginary parts), which is exactly $R(\alpha)$ acting on the flag factor by Proposition 19.4. $\square$

Example 19.7 (The composition obstruction, worked explicitly). The complex tensor product satisfies $(cv) \otimes w = v \otimes (cw)$ for any scalar c : a phase can be moved between factors without changing the product. Take $c = i$, $v = w = |0\rangle$: $i|00\rangle$ has two equally valid product decompositions, $(i|0\rangle) \otimes |0\rangle$ and $|0\rangle \otimes (i|0\rangle)$. Applying $\mathcal{S}$ to each local factor before taking a (real) tensor product gives, for the first decomposition, $\mathcal{S}(i|0\rangle) \otimes \mathcal{S}(|0\rangle) = (|0\rangle \otimes F_1) \otimes (|0\rangle \otimes F_0)$, and for the

second, $\mathcal{S}(|0\rangle) \otimes \mathcal{S}(i|0\rangle) = (|0\rangle \otimes F_0) \otimes (|0\rangle \otimes F_1)$ — two different vectors in $\mathbb{R}^4 \otimes \mathbb{R}^2 \otimes \mathbb{R}^4 \otimes \mathbb{R}^2$ (16 real dimensions), both claiming to represent the single vector $i|00\rangle$, which the correct global map $\mathcal{S}$ applied directly to the 4-dimensional composite state sends to one well-defined vector in $\mathbb{R}^4 \otimes \mathbb{R}^2$ (8 real dimensions). The naive local construction is not merely inefficient by a factor of two; it is not well-defined at all, since it depends on an arbitrary choice of which factor “carries” the phase.

Definition 19.8 (Admissibility equivalence). On the raw local-flag tensor product, define $x \sim_A x'$ if x, x' produce identical statistics under every measurement built from local operators of the form $\mathcal{T}(\Pi) = \text{Re}(\Pi) \otimes I_F + \text{Im}(\Pi) \otimes J_F$ (Proposition 19.5 identifies J_F as the flag realization of i).

Proposition 19.9. $\sim_A$ is an equivalence relation.

Proof. Reflexivity and symmetry are immediate from the definition (agreement of statistics is symmetric and self-satisfied). Transitivity: if $x \sim_A y$ and $y \sim_A z$, every measurement’s statistics agree on x, y and on y, z , hence on x, z . $\square$

Quotienting the raw local-flag space by $\sim_A$ removes exactly the redundancy exhibited in Example 19.7 — the two decompositions of $i|00\rangle$ become a single equivalence class — while preserving every physical prediction, since $\sim_A$ is defined to preserve exactly the statistics that matter.

Proposition 19.10 (Quotienting reorganizes ambiguity, it does not remove it). Let a group G act on a space X , with quotient map $\pi : X \rightarrow X/G$. The freedom to choose a representative $x \in \pi^{-1}([x])$ is exactly the G -orbit of x : quotienting removes the need to choose a representative without destroying the freedom itself.

Proof. By definition $\pi^{-1}([x]) = \{gx : g \in G\}$, the orbit of x . Passing to X/G replaces a choice of representative with a single point $[x]$, but the fiber over $[x]$ is exactly the orbit — the same set of gauge-related representatives as before, now organized as a fiber rather than presented as ambiguity in a single vector. $\square$

Remark 19.11. Applied to Example 19.7: the ambiguity in which local factor carries a global phase does not disappear under the quotient. It becomes the orbit structure of the flag-rotation group acting on the raw local tensor product — the same information, differently organized. This is the composite-system mechanism underlying Barrios Hita et al.’s construction [11], derived here from Definition 19.8 and Propositions 19.9–19.10 rather than imported from it.

Ledger 19.12 (End of Chapter 19). Established: the amplitwist group structure and its identification with complex multiplication (Theorems/Propositions 19.2–19.5), fully self-contained; the composite-system phase-redistribution obstruction, illustrated with a freshly derived and independently verified minimal example (Example 19.7) rather than imported formulas; the admissibility-equivalence quotient construction and the general fact that quotienting reorganizes rather than destroys ambiguity (Propositions 19.9–19.10).

Not established: that this quotient construction, applied to general n -partite systems, reproduces every prediction of standard quantum mechanics — that is the actual technical content of Barrios Hita et al.’s Letter [11], a substantially harder claim this chapter’s minimal two-factor illustration does not attempt to establish. The elementary rotation-scaling content of this chapter (§19.1–19.3) predates quantum mechanics entirely and should not be mistaken for that harder result.

Part IV

Deepening the Reconstruction

Chapter 20

Minimal Witnesses in Higher Dimensions

Parts I through III extended the reconstruction toward standard quantum mechanics, chapter by chapter, until Chapter 18 closed that arc. This part does something different: it no longer moves toward recovering more of the textbook formalism, but stress-tests what has already been built, asking how far the same admissibility machinery pushes on its own terms. The governing questions become "how little is actually enough?" and "where does this specific reconstruction genuinely run out?" — a research programme extending the argument, not an appendix collecting loose ends.

Chapter 7 left open what $m_\beta(\mathbb{R}, \mathbb{C}; d)$ is for $d \geq 3$, noting that Chapter 3's $n \geq 3$ incompleteness theorem gives a single-matrix separation whose relation to the graph-level minimal witness was unresolved. This chapter resolves it — and the answer inverts the $d = 2$ picture rather than extending it.

Theorem 20.1 (Collapse of the minimal witness for $d \geq 3$). For every $d \geq 3$, $m_\beta(\mathbb{R}, \mathbb{C}; d) = 0$: a single edge, carrying no cycle at all, already witnesses $\mathcal{R}_\mathbb{R}(G, d) \subsetneq \mathcal{R}_\mathbb{C}(G, d)$. This is conditional on nothing beyond Definitions 7.1–7.2 as already stated; no new assumption is introduced.

Proof. Let P_2 be the two-vertex, one-edge graph ($\beta_1(P_2) = 1 - 2 + 1 = 0$). By Theorem 3.3 (the $n \geq 3$ incompleteness theorem), there is a unistochastic $\Gamma \in \text{US}(3) \setminus \text{OS}(3)$ — the modulus-square of the 3×3 DFT matrix. Label the single edge of P_2 with Γ . Since $\Gamma \notin \text{OS}(3)$, no real frame assignment reproduces it: this edge-labeling is not in $\mathcal{R}_\mathbb{R}(P_2, 3)$ (indeed it fails even edgewise real-admissibility, and by Proposition 5.6 — global realizability on a tree is equivalent to edgewise admissibility — failing the latter is exactly failing the former). Since $\Gamma \in \text{US}(3)$ by construction, it is realizable over $\mathbb{C}$: the labeling is in $\mathcal{R}_\mathbb{C}(P_2, 3)$. Hence P_2 is a witness with $\beta_1(P_2) = 0$, giving $m_\beta(\mathbb{R}, \mathbb{C}; 3) \leq 0$, and $\beta_1(G) \geq 0$ for every graph, so $m_\beta(\mathbb{R}, \mathbb{C}; 3) = 0$ exactly. The same Γ , embedded as one edge of any tree on $d = 3$ contexts for $d \geq 3$ (padding the unused dimensions trivially), gives the identical argument for every $d \geq 3$. $\square$

Remark 20.2 (Why this inverts, rather than extends, the $d = 2$ result). Chapter 7's proof that $m_\beta(\mathbb{R}, \mathbb{C}; 2) \geq 1$ used Proposition 5.6 together with a fact specific to $d = 2$: $\text{US}(2) = \text{OS}(2)$ (Theorem 3.1), so every edge individually admissible over $\mathbb{C}$ at $d = 2$ is automatically admissible over $\mathbb{R}$ too, and no tree can separate the fields. That equality is exactly what Theorem 3.3 shows fails at $d = 3$. The $d = 2$ case is the exceptional one: cycles are needed there because single edges cannot yet distinguish the fields. From $d = 3$ onward, single edges already can, and the graph-theoretic machinery of Chapters 5–7 — cycle rank, spanning trees, the triangle-gluing theorem — is not needed to separate $\mathbb{R}$ from $\mathbb{C}$ at all. It remains exactly what is needed for the sharper, narrower question that motivated it: whether individually admissible data (data that passes every edge test) can still fail to glue globally. That question is nontrivial at every $d \geq 2$, and is not addressed by Theorem 20.1.

Remark 20.3 (What Theorem 20.1 does not say). Theorem 20.1 establishes a statement about a minimum:

$$\min\{\text{witness size}\} = \begin{cases} 3, & d = 2, \\ 1, & d \geq 3. \end{cases} \quad (20.1)$$

It does not establish that graph-theoretic gluing obstructions disappear for $d \geq 3$, and should not be read that way. The question left open is whether there exist edgewise-admissible labelings on cyclic graphs — data that individually passes every single-edge test, so that no Theorem 20.1-style witness is available — that nevertheless fail to admit a global realization, generalizing the $d = 2$ MUB triangle phenomenon of Chapter 6 to $d \geq 3$. This is a logically independent question from the one this chapter answers: the cheaper single-edge obstruction found here does not preclude a separate, more expensive cycle-level obstruction also existing alongside it. If future work shows no such edgewise-admissible-but- globally-inadmissible examples exist at $d \geq 3$, Theorem 20.1 would then collapse the entire obstruction theory to single-edge tests. If such examples are found instead, the theory acquires two independent layers — local (single-edge) and global (cycle-gluing) obstructions — the way many areas of mathematics distinguish satisfying every local constraint from admitting a global solution. Neither possibility is resolved here.

Ledger 20.4 (End of Chapter 20). Established: $m_\beta(\mathbb{R}, \mathbb{C}; d) = 0$ for all $d \geq 3$ (Theorem 20.1), resolving Chapter 7’s first open question; the reason is identified precisely as the failure of $\text{US}(d) = \text{OS}(d)$ beyond $d = 2$, not a new phenomenon requiring new machinery.

Not established: anything about $m_\beta(\mathbb{C}, \mathbb{H}; d)$ (quaternionic separation, Chapter 7’s second open question, addressed only partially in Chapter 21); the density-threshold question, still untouched (Chapter 7’s fourth open question resolves neither here nor in Chapter 22, which settles the forbidden-subgraph question at $d = 2$ but not the density question at any d). The cycle-based machinery of Chapters 5–7 remains the right tool for the gluing question it was built for, and this chapter’s result should not be read as making that machinery unnecessary — only as showing it answers a different, narrower question than “does $\mathbb{R}$ ever fail,” which single edges already settle from $d = 3$ on (see Remark 20.3 above for exactly what remains open about that narrower question).

Remark 20.5 (Two distinct hazards, not one). This chapter and Chapter 18’s discussion of recent real-valued quantum mechanics literature share a family resemblance worth naming precisely rather than merging. Both are cases where something that looked like a fundamental obstruction turned out to be contingent — but on different kinds of contingency, and the difference matters for what a reader should check.

The 2025–2026 papers concern representation-dependence: the Kronecker product and Hofreumon and Woods’ $\otimes_r$ both describe the identical physical composition, and the apparent obstruction to real quantum theory belonged to a choice of notation, not to the physics. The diagnostic is recoordinatization: does the obstruction survive a change of representation for the same object?

Theorem 20.1 concerns a different hazard: a low-dimensional coincidence. $\text{US}(2) = \text{OS}(2)$ is not a notational artifact of how 2×2 matrices happen to be written; it is a true, dimension-specific fact that is simply false one dimension higher. No change of representation at $d = 2$ would have revealed this — the only diagnostic is varying the parameter (d) and recomputing, which is a genuinely different check than asking whether a result survives recoordinatization.

Both hazards share a single moral — a pattern that holds in a special case can masquerade as the general rule if the special case is never stress-tested against the parameter, or representation, that made it special — but the stress test differs by hazard, and treating them as interchangeable would leave a reader checking only one of the two ways a result can be an artifact rather than a fact.

Chapter 21

Toward Quaternionic Separation

Chapter 7’s second open question, $m_\beta(\mathbb{C}, \mathbb{H}; d)$, was left entirely untouched pending the composite-system chapters. Those chapters (Ch. 8–12) found quaternionic composition genuinely obstructed — the naive product of local quaternionic Hermitian bases is already overcomplete (Ch. 9’s remark). This chapter asks a narrower question first: not composite systems, but whether the single-system graph-realizability machinery of Chapters 5–7 — which never required composing separate systems, only composing frame transitions of one system along a graph — extends to $\mathbb{H}$ at all.

21.1 What Generalizes, and Why

The graph-realizability machinery (Proposition 5.6, Theorem 5.7) was built entirely from matrix multiplication of frame transitions: $F_v := F_u L_{uv}$ propagated along a tree, and cycle closure $L_{A_0 A_1} \cdots L_{A_{k-1} A_0} = I$ via telescoping. Neither proof used commutativity anywhere — only associativity, and the fact that $\pi(L)_{ij}$ (the modulus-squared of a matrix entry) is well-defined and nonnegative.

Proposition 21.1 (The machinery is field-agnostic). Proposition 5.6 and Theorem 5.7 hold verbatim with $\mathbb{R}$ replaced by $\mathbb{H}$ throughout, where $\pi(L)_{ij} := |L_{ij}|_{\mathbb{H}}^2 = L_{ij} \overline{L_{ij}}$ (the quaternionic norm, a nonnegative real number for every quaternion $L_{ij} = a + bi + cj + dk$), and ”unitary” means $L^\dagger L = I$ with $\dagger$ the quaternionic conjugate transpose.

Proof. Quaternionic matrix multiplication is associative (inherited from associativity of quaternion multiplication itself) though not commutative. Every step of Proposition 5.6’s proof (unique tree-path propagation) and Theorem 5.7’s proof (construct from a tree-lift choice, check non-tree edges) used only associativity and the definition of π ; neither invoked commutativity of matrix entries or of the underlying scalars. $\square$

This was checked directly, not merely asserted: quaternionic associativity, a genuine quaternionic unitary matrix (via quaternionic Gram–Schmidt) satisfying $U^\dagger U = I$, exact cycle closure $L_{ZX} L_{XY} L_{YZ} = I$ for random quaternionic frames, and exact reproduction of tree-edge data under propagation, all confirmed numerically before this proposition was written.

21.2 The $\mathbb{C}$ – $\mathbb{H}$ Minimal Witness Question

Definitions 7.1–7.2 are already stated for general $\mathbb{F}$, so $m_\beta(\mathbb{C}, \mathbb{H}; d)$ is well-posed immediately given Proposition 21.1. Two facts constrain where a witness could live.

Proposition 21.2 ($\mathbb{C} \subset \mathbb{H}$ as an inclusion of admissible sets). $\mathcal{R}_{\mathbb{C}}(G, d) \subseteq \mathcal{R}_{\mathbb{H}}(G, d)$ for every G, d .

Proof. $\mathbb{C}$ embeds as a subalgebra of $\mathbb{H}$ ($a + bi \mapsto a + bi + 0j + 0k$), so every complex unitary matrix is a quaternionic unitary matrix with entries restricted to this subalgebra; any real frame assignment achieving a labeling over $\mathbb{C}$ achieves it over $\mathbb{H}$ using the identical matrices. $\square$

Proposition 21.3 (No single-edge witness at $d = 2$). $\mathcal{R}_{\mathbb{R}}(G, 2) = \mathcal{R}_{\mathbb{C}}(G, 2) = \mathcal{R}_{\mathbb{H}}(G, 2)$ for every tree G ; in particular $m_{\beta}(\mathbb{C}, \mathbb{H}; 2) \geq 1$, if it exists at all.

Proof. Theorem 3.1 shows every $B \in \text{DS}(2)$ is already orthostochastic: $\text{OS}(2) = \text{DS}(2)$. Since $\mathbb{R} \subset \mathbb{C} \subset \mathbb{H}$, the same real orthogonal matrix witnessing $\mathbb{R}$ -admissibility witnesses $\mathbb{C}$ - and $\mathbb{H}$ -admissibility too, so all three edgewise admissible sets coincide at $d = 2$, and by Proposition 5.6 (both real and quaternionic versions) no tree can separate any pair of the three fields at $d = 2$. $\square$

21.3 What This Chapter Does Not Establish

Unlike Chapter 20's resolution of the real/complex question at $d \geq 3$, no analogous collapse is established here. That resolution rested on an explicit, verified single matrix (Theorem 3.3's DFT-derived $\Gamma \in \text{US}(3) \setminus \text{OS}(3)$) exhibiting the separation directly. The analogous question for $\mathbb{C}$ versus $\mathbb{H}$ — does there exist, at some small d , a quaternionic-unistochastic matrix admitting no complex realization — has no such explicit witness in hand. Constructing one requires either an explicit small-dimension quaternionic unitary matrix whose modulus-squared provably escapes every complex realization (the direct analogue of the DFT construction, but genuinely harder: the relevant non-existence fact would concern complex, not real, matrix constraints, and no argument as elementary as the Hadamard-matrix non-existence check of Theorem 3.3 is currently in hand), or a correct quaternionic generalization of the mutually-unbiased-basis counting bound that underlies Chapter 6's triangle argument — and the standard derivation of that bound leans on structural facts about the complex traceless-Hermitian-operator space that do not obviously carry over: a quaternionic orthonormal basis carries $\text{Sp}(1)$ (three real parameters) of local phase freedom per vector rather than the complex case's $U(1)$ (one real parameter), which changes the dimension count in a way this book has not worked out correctly. Rather than risk asserting an unverified bound, this is left as the precise, well-posed target for future work.

Remark 21.4 (The gap, isolated). Propositions 21.2 and 21.3 narrow $m_{\beta}(\mathbb{C}, \mathbb{H}; d)$, if it exists at all, to $d \geq 3$. A separation at some such d would follow from either:

- (i) an explicit bistochastic matrix on some small $d \geq 3$ realizable quaternionically but not complexly — the direct analogue of Theorem 3.3's DFT construction, requiring a non-existence fact about complex matrices in place of that theorem's non-existence of a real Hadamard matrix; or
- (ii) a structural obstruction — most plausibly a corrected quaternionic mutually-unbiased-basis counting bound, redone from first principles to account for the $\text{Sp}(1)$ (rather than $U(1)$) local phase freedom per basis vector — proving such a matrix must exist without exhibiting one directly, in the style of Chapter 6's cycle argument.

Neither is completed here. What this chapter contributes is the reduction: the search is not open at every d , only at $d \geq 3$, and the machinery needed to verify a candidate witness, once found, is already in place (Proposition 21.1).

Ledger 21.5 (End of Chapter 21). Established: the graph-realizability machinery of Chapters 5–7 generalizes to $\mathbb{H}$ verbatim, verified computationally (Proposition 21.1); $\mathcal{R}_{\mathbb{C}} \subseteq \mathcal{R}_{\mathbb{H}}$ always (Proposition 21.2); no single-edge witness exists at $d = 2$ for any pairing of $\mathbb{R}, \mathbb{C}, \mathbb{H}$ (Proposition 21.3).

Not established: $m_\beta(\mathbb{C}, \mathbb{H}; d)$ for any d — neither an explicit witness nor a correct counting bound. This chapter narrows the search (rules out $d = 2$ single edges, confirms the machinery needed to search $d \geq 3$ is sound) without completing it. The quaternionic MUB-counting question, and the composite-system obstruction flagged in Chapter 9, remain the two live approaches.

Chapter 22

The Forbidden Subgraph, Resolved at $d = 2$

Chapter 7’s third open question asked for a finite family of forbidden context subgraphs characterizing real-realizability. This chapter answers it completely at $d = 2$: the forbidden family has exactly one member, a bare cycle, and the characterization is that G is always real-realizable if and only if G is a forest.

22.1 Subdivision Preserves the Obstruction

Lemma 22.1 (Subdivision lemma). Let G' be obtained from a context graph G by subdividing an edge (u, v) — replacing it with a path $u, w_1, \dots, w_k, v$ through new vertices — and labeling every new edge with canonical angle 0. Then G is realizable with (u, v) labeled θ , together with the rest of G ’s labeling, if and only if G' is realizable with the same labeling on the unchanged edges and every subdivision edge labeled 0.

Proof. A canonical angle of 0 on an edge (w_{i-1}, w_i) is achieved exactly by the real frame choice $a_{w_i} = a_{w_{i-1}}$ (one valid branch of the fibre described in §5.3). Propagating this choice along the entire subdivision path forces $a_v = a_u$ exactly, identically to what a direct edge (u, v) with canonical angle 0 would force. For any other target angle θ on the original edge (u, v) , choose instead $a_{w_1} = \dots = a_{w_k} = a_u$ (still trivial edges) and require $a_v - a_u$ to realize θ exactly as in the unsubdivided graph — the trivial path contributes nothing to the relationship between a_u and a_v , so realizability of G with $(u, v) = \theta$ and realizability of G' with the subdivision (trivial edges) plus (u, v) ’s role now played by the path’s endpoints are the same condition on the same free parameters $\{a_x\}_{x \in V(G)}$, with the new vertices’ angles pinned to a_u and contributing no additional freedom or constraint. $\square$

Corollary 22.2 (Every cycle length inherits the triangle obstruction). For every $n \geq 3$, there is an assignment of canonical angles to the edges of the n -cycle C_n , individually admissible on every edge, that is not globally real-realizable.

Proof. C_n is a subdivision of C_3 for every $n \geq 3$ (subdivide $n - 3$ of the triangle’s edges, distributed arbitrarily, to reach length n). Apply Lemma 22.1 to the mutually-unbiased triangle labeling of Corollary 6.4 (canonical angle $\pi/4$ on all three original edges), labeling every subdivision edge 0: by Corollary 6.4 the triangle labeling is unrealizable, so by the lemma the resulting C_n labeling is unrealizable too. $\square$

This was checked computationally before being stated as a corollary: exact branch-search confirmed the induced labeling is realizable on neither C_4 nor C_5 (zero solutions found over all $4^3 = 64$ and $4^4 = 256$ branch combinations respectively), while a control labeling differing only in the closing edge (replacing the inherited $\pi/4$ with the trivially-realizable 0) is realizable on both.

22.2 The Forbidden Subgraph Theorem

Theorem 22.3 (Forbidden subgraph, $d = 2$). $\mathcal{R}_{\mathbb{R}}(G, 2) = \mathcal{D}(G, 2)$ (every individually admissible edge-labeling of G is globally real-realizable) if and only if G is a forest.

Proof. ($\Leftarrow$) If G is a forest, apply Proposition 5.6 to each connected component: any individually admissible labeling extends to a global real frame assignment on each tree, hence on all of G .

($\Rightarrow$) If G is not a forest, G contains a cycle (a graph with $|E| \geq |V|$ on some connected component must contain one, since a forest has at most $|V| - 1$ edges per component); let C_n , $n \geq 3$, be that cycle, viewed as a subgraph of G . Apply Corollary 22.2 to label C_n 's edges with an individually-admissible, globally-unrealizable pattern, and label every edge of G outside C_n with canonical angle 0 (trivially admissible). The resulting labeling of all of G is individually admissible on every edge; it cannot be globally realized, since any global realization would in particular realize C_n 's edges, contradicting Corollary 22.2. Hence $\mathcal{D}(G, 2) \not\subseteq \mathcal{R}_{\mathbb{R}}(G, 2)$. $\square$

Remark 22.4. This is the forbidden-subgraph family in its simplest possible form: a single forbidden pattern, "any cycle," rather than a genuinely infinite or complicated family. Equivalently: $\mathcal{F} = \{C_3, C_4, C_5, \dots\}$ works, but by Corollary 22.2 it can be stated purely topologically as $\mathcal{F} = \{\text{any cycle}\}$, i.e. G is forbidden-subgraph-free exactly when $\beta_1(G) = 0$.

22.3 Contrast with $d \geq 3$

Remark 22.5. Theorem 22.3 is specific to $d = 2$, and for the same reason Chapter 20's collapse was specific to $d \geq 3$: the proof's forward direction rests on Proposition 5.6 making trees exactly safe, which in turn requires that individually-admissible edge data always extends — true at $d = 2$ only because $\text{US}(2) = \text{OS}(2)$ (Theorem 3.1) makes edgewise admissibility the same condition over $\mathbb{R}$ and $\mathbb{C}$. At $d \geq 3$, Chapter 20 showed a single edge (a tree with one edge, $\beta_1 = 0$) can already carry data admissible over $\mathbb{C}$ but not $\mathbb{R}$ (Theorem 3.3's DFT example). So at $d \geq 3$, the topological condition "G is a forest" is not sufficient for $\mathcal{R}_{\mathbb{R}}(G, d) = \mathcal{D}(G, d)$ — trees can already fail, so the $d = 2$ forbidden-subgraph theorem does not generalize by simply keeping the same statement. What replaces it at $d \geq 3$ is not a purely topological condition at all, but the data-dependent single-edge obstruction of Chapter 20: the relevant "forbidden pattern" becomes a specific matrix (or class of matrices, per Chapter 3's open question about whether every non-orthostochastic unistochastic matrix fails for the same reason), not a graph shape.

Ledger 22.6 (End of Chapter 22). Established: the forbidden-subgraph question of Chapter 7 is fully resolved at $d = 2$ (Theorem 22.3): G is always real-realizable if and only if it is a forest, via a subdivision lemma proved algebraically and cross-checked computationally at two cycle lengths.

Not established: any analogous purely topological characterization at $d \geq 3$, where the obstruction is data-dependent even on trees (Chapter 20); the density-threshold question of Chapter 7 remains entirely untouched and would require a probability measure on edge-label data this book still has not defined.

Chapter 23

The Density Question, Resolved Sharply

Chapter 7’s fourth open question asked whether, for a random context graph, the probability that admissible edge data requires $\mathbf{C}$ approaches 1 as edge density increases — an Erdős–Rényi-style asymptotic threshold, explicitly flagged there as needing a probability measure on edge-label data this book had not yet defined. This chapter defines that measure and answers the question, but the answer is sharper than the question anticipated: there is no asymptotic threshold to find, because the threshold is already crossed at the first cycle.

23.1 A Natural Measure, and a Correction

Definition 23.1 (Independent continuous edge measure). For a context graph G at $d = 2$, equip each edge with an independent $\text{Uniform}[0, \pi/4]$ distribution over its canonical angle, independently across edges.

This is the natural candidate the question calls for: no correlation between edges is assumed, and no edge value is privileged.

Before proceeding, a claim made in passing in Chapter 6 needs correcting. That chapter’s discussion of the triangle-gluing theorem stated that “generic triples of edge angles do admit real joint realizations.” Checked directly against Definition 23.1, this is false, and has been corrected at the point it was made (Chapter 6’s remark following Theorem 6.2): realizability is not the generic case. It is the exception.

23.2 The Sharp Result

Theorem 23.2 (Realizability is measure-zero once $\beta_1(G) \geq 1$). For any context graph G with $\beta_1(G) \geq 1$, under Definition 23.1, $\Pr[\text{edge data is globally real-realizable}] = 0$.

Proof. Fix a spanning tree T and the $\beta_1(G)$ non-tree edges. By Theorem 5.7, global realizability requires a single choice of tree-edge lifts under which every non-tree edge matches its target angle. For fixed tree-edge angles, each non-tree edge’s target must equal one of finitely many values determined by the tree data (Theorem 6.2 for each fundamental cycle) — a codimension-one condition on that edge’s independently-drawn value. With $\beta_1(G)$ non-tree edges drawn independently, satisfying all $\beta_1(G) \geq 1$ such conditions simultaneously is a codimension- $\beta_1(G)$ event under a continuous product measure, hence has probability zero. $\square$

This was verified computationally before being stated as a theorem, at two different values of β_1 : 0.05% of 2,000,000 fully independent triangle trials matched realizability within a generous numerical tolerance of 10^{-4} (consistent with probability exactly zero as the tolerance shrinks), and 0 of 200,000 trials matched on K_4 ($\beta_1 = 3$), the higher codimension making the event correspondingly harder to hit even at finite tolerance.

23.3 Why This Is Not an Erdős–Rényi Threshold

The question as posed in Chapter 7 anticipated an asymptotic phenomenon: a critical edge density below which real realizability is typical and above which it is not, in the manner of the giant-component threshold. Theorem 23.2 shows this picture does not apply. There is no gradual transition and no critical density to locate: the moment a graph contains even one cycle ($\beta_1(G) = 1$, the minimal case, achieved by a bare triangle), generic edge data already fails to realize over $\mathbb{R}$ with probability one. Adding further edges (increasing $\beta_1(G)$ further) does not create a threshold that was previously absent; it was already crossed.

Remark 23.3. This should be read alongside Chapter 20’s inversion result in the same spirit: an intuition imported from a different context (the Erdős–Rényi threshold, or “higher dimension means larger witnesses”) turns out not to describe what actually happens here, and the correction is sharper and more interesting than the naive expectation. Where Chapter 20 found the minimal witness shrinks rather than grows with dimension, this chapter finds the density threshold degenerates to a single point ($\beta_1 = 1$) rather than sweeping out a nontrivial range.

Ledger 23.4 (End of Chapter 23). Established: the density-threshold question of Chapter 7 is resolved (Theorem 23.2): under the natural independent-continuous edge measure, real realizability has probability zero for every graph with $\beta_1(G) \geq 1$, not an asymptotic phenomenon requiring a critical density. A false claim in Chapter 6 (“generic triples do admit real realizations”) is corrected at its source as part of this resolution.

Not established: anything about non-continuous or correlated measures on edge data, which might in principle behave differently (e.g. a measure concentrated on real-derived triples would trivially give probability one, the opposite regime); anything about $d \geq 3$, where Theorem 23.2’s proof — via the fundamental-cycle characterization of Theorem 5.7 — still applies verbatim to the graph-level obstruction, but where Chapter 20’s cheaper single-edge obstruction is a separate phenomenon this chapter’s measure has not been extended to analyze.

Chapter 24

A Quaternionic Mutually-Unbiased-Basis Bound

Chapter 21's closing remark identified two routes to a $\mathbb{C}$ – $\mathbb{H}$ separation witness: an explicit matrix construction, or a corrected quaternionic mutually-unbiased-basis counting bound. It flagged that the standard derivation leans on structural facts not yet worked out for $\mathbb{H}$ — specifically the $\mathrm{Sp}(1)$ versus $U(1)$ local phase freedom. This chapter works that derivation through carefully, checking each step computationally before trusting it, and arrives at a genuine bound: quaternionic mutually unbiased bases can outnumber complex ones by roughly a factor of two.

24.1 Two Traps, and Why Neither Bites

The standard complex MUB bound rests on two facts: (i) the d traceless projectors of one orthonormal basis span a $(d - 1)$ -dimensional subspace of the ambient traceless Hermitian space, and (ii) mutually unbiased bases give orthogonal such subspaces, via $\mathrm{Tr}[(P_i - I/d)(Q_j - I/d)] = \mathrm{Tr}(P_i Q_j) - 1/d = 1/d - 1/d = 0$. Generalizing to $\mathbb{H}$ threatens both facts, for different reasons.

Proposition 24.1 (Fact (i) survives $\mathrm{Sp}(1)$ phase freedom). For a quaternionic orthonormal basis $\{|e_i\rangle\}_{i=1}^d$, the projector $P_i = |e_i\rangle\langle e_i|$ is independent of which unit quaternion $q \in \mathrm{Sp}(1)$ represents the same ray: $|e_i\rangle q$ gives the identical projector.

Proof. $(|e_i\rangle q)(|e_i\rangle q)^\dagger = |e_i\rangle q \bar{q} \langle e_i| = |e_i\rangle |q|^2 \langle e_i| = |e_i\rangle \langle e_i|$, using $(Aq)^\dagger = \bar{q} A^\dagger$ and $|q|^2 = 1$ for unit quaternions. $\square$

The three-fold $\mathrm{Sp}(1)$ freedom in choosing a ray representative therefore contributes nothing extra to the space spanned by $\{P_i - I/d\}$: it is still exactly d objects, still spanning at most a $(d - 1)$ -dimensional subspace (one linear relation, $\sum_i (P_i - I/d) = 0$). This was confirmed computationally: for a random quaternionic orthonormal basis at $d = 3$, the real rank of the span of the three traceless projectors is exactly 2.

Proposition 24.2 (Fact (ii) requires the correct inner product). Quaternionic matrix trace is not cyclic in general: $\mathrm{Tr}(AB) \neq \mathrm{Tr}(BA)$. However $\mathrm{Re}[\mathrm{Tr}(AB)] = \mathrm{Re}[\mathrm{Tr}(BA)]$ always, and for rank-one Hermitian projectors, $\mathrm{Re}[\mathrm{Tr}(P_i Q_j)] = |\langle e_i | f_j \rangle|_{\mathbb{H}}^2$.

This was checked computationally, not merely asserted: for random quaternionic matrices A, B , $\mathrm{Tr}(AB) \neq \mathrm{Tr}(BA)$ as full quaternions, but their real parts agree exactly; for projectors built from random quaternionic unit vectors, $\mathrm{Re}[\mathrm{Tr}(P_0 Q_0)]$ matches $|\langle e_0 | f_0 \rangle|_{\mathbb{H}}^2$ to numerical precision, and the identity $\mathrm{Re}[\mathrm{Tr}(P_i Q_j)] - 1/d = \langle P_i - I/d, Q_j - I/d \rangle_{\mathbb{H}}$ (using $\langle A, B \rangle_{\mathbb{H}} := \mathrm{Re}[\mathrm{Tr}(A^\dagger B)]$ as the real inner product on the quaternionic Hermitian matrix space) holds exactly. This is the resolution: $\mathrm{Re}[\mathrm{Tr}(\cdot)]$, not the raw quaternionic trace, is the correct symmetric bilinear form, and fact (ii) survives once stated in these terms.

24.2 The Bound

Theorem 24.3 (Quaternionic MUB bound). For m pairwise mutually unbiased orthonormal bases of $\mathbb{H}^d$,

$$m \leq 2d + 1. \quad (24.1)$$

Proof. By Proposition 24.1, each basis contributes a $(d - 1)$ -dimensional real subspace of the $(\beta_{\mathbb{H}}(d) - 1)$ -dimensional traceless quaternionic Hermitian space (Chapter 8's $\beta_{\mathbb{H}}(d) = d(2d - 1)$). By Proposition 24.2, mutual unbiasedness ($\text{Re}[\text{Tr}(P_i Q_j)] = 1/d$ for all i, j) gives $\langle P_i - I/d, Q_j - I/d \rangle_{\mathbb{H}} = 0$: the subspaces from distinct mutually unbiased bases are pairwise orthogonal. Hence m such subspaces fit inside the ambient space only if $m(d - 1) \leq \beta_{\mathbb{H}}(d) - 1 = d(2d - 1) - 1 = (2d + 1)(d - 1)$, giving $m \leq 2d + 1$. $\square$

Remark 24.4 (Validation against the cited real bound). Applying the identical method with $\beta_{\mathbb{R}}(d) = d(d + 1)/2$ in place of $\beta_{\mathbb{H}}(d)$ gives $m \leq (\beta_{\mathbb{R}}(d) - 1)/(d - 1) = d/2 + 1$ — exactly the $N_{\mathbb{R}}(d) \leq \lfloor d/2 \rfloor + 1$ bound of Boykin, Sitharam, Tarifi, and Wocjan already cited in Chapter 6, before flooring for integer d . The same method applied with $\beta_{\mathbb{C}}(d) = d^2$ gives the standard $m \leq d + 1$. Reproducing a trusted external result via the identical derivation, before trusting the new quaternionic case, is the check this chapter's result was built to survive.

24.3 What This Bound Does and Does Not Establish

Theorem 24.3 is an upper bound. It shows quaternionic mutually unbiased bases could outnumber complex ones by roughly a factor of two ($2d + 1$ versus $d + 1$) — it does not show that they actually do. Two further steps, neither attempted here, would be needed for an explicit $\mathbb{C}$ – $\mathbb{H}$ separation witness in the sense of Chapter 7:

- (a) Existence. Does there exist a dimension d and an explicit set of more than $d + 1$ pairwise mutually unbiased quaternionic bases of $\mathbb{H}^d$? Theorem 24.3 permits this in principle (the bound is not obviously tight, but it does not forbid exceeding $d + 1$); no construction is given here.
- (b) Translation to graph realizability. Even given such a construction, exhibiting a genuine witness in the sense of Chapter 7 requires showing the resulting transition data is individually $\mathbb{C}$ -admissible on every edge yet fails to glue over $\mathbb{C}$ globally — the direct quaternionic analogue of Chapter 6's triangle argument, which was worked out in the $d = 2$ real case using the specific angle machinery of §5.3, not yet generalized to general d and general fields.

Ledger 24.5 (End of Chapter 24). Established: a quaternionic MUB bound, $m \leq 2d + 1$ (Theorem 24.3), derived from first principles with every step checked computationally rather than assumed — the $\text{Sp}(1)$ phase-freedom concern resolves harmlessly (Proposition 24.1), and the correct symmetric inner product ($\text{Re}[\text{Tr}(\cdot)]$, not the raw quaternionic trace) makes the orthogonality argument go through (Proposition 24.2). The method is validated by exactly reproducing the cited real-MUB bound when applied to $\mathbb{R}$ instead of $\mathbb{H}$.

Not established: whether quaternionic MUB sets actually exceeding $d + 1$ exist at any d (an existence question, not addressed by an upper bound); whether such a set, if found, translates into an actual graph- realizability separation witness in the sense of Chapter 7. This chapter narrows Chapter 21's open dichotomy toward route (ii) — a structural bound is now in hand — without completing it.

Appendix A

Notation and Conventions

This appendix collects, in one place, notation used repeatedly across the book without being redefined at each occurrence. It introduces nothing new; every item below is defined at its first occurrence in the main text, cross-referenced here for convenience.

A.1 Number Systems

$\mathbb{R} \subset \mathbb{C} \subset \mathbb{H}$: real numbers, complex numbers, quaternions, in the inclusions used throughout Part II onward (Chapter 21, Proposition 21.2).

$\mathbb{F}$: a generic placeholder for one of $\mathbb{R}, \mathbb{C}, \mathbb{H}$, used whenever a definition or theorem is stated uniformly across all three.

A.2 Matrix Classes

$\text{DS}(n)$: $n \times n$ doubly (bi-)stochastic matrices — nonnegative entries, all row and column sums equal to 1 (Chapter 2).

$\text{OS}(n)$, $\text{US}(n)$: orthostochastic andunistochastic matrices — $B_{ij} = O_{ij}^2$ for real orthogonal O , or $B_{ij} = |U_{ij}|^2$ for unitary U , respectively (Chapter 3). $\text{OS}(n) \subseteq \text{US}(n) \subseteq \text{DS}(n)$ always; equality of the first two holds only at $n = 2$ (Theorem 3.1) and fails from $n = 3$ on (Theorem 3.3).

$\text{Herm}_d(\mathbb{F})$: $d \times d$ Hermitian matrices over $\mathbb{F}$ (symmetric for $\mathbb{R}$, conjugate-symmetric for $\mathbb{C}$, quaternionic-conjugate-symmetric for $\mathbb{H}$); $\beta_{\mathbb{F}}(d) := \dim_{\mathbb{R}} \text{Herm}_d(\mathbb{F})$, with $\beta_{\mathbb{R}}(d) = d(d+1)/2$, $\beta_{\mathbb{C}}(d) = d^2$, $\beta_{\mathbb{H}}(d) = d(2d-1)$ (Chapter 8, Proposition 8.1).

$\mathcal{D}_d(\mathbb{F})$: the physical state space, positive semidefinite trace-one elements of $\text{Herm}_d(\mathbb{F})$ (Chapter 10).

A.3 The Canonical Angle (Real 2×2 Case)

$\theta \in [0, \pi/4]$: the canonical unsigned transition angle between two real two-dimensional contexts, defined via $B(\theta)_{ii} = \cos^2 \theta$, unique after quotienting the gauge and projection-loss identifications of §5.3 (Proposition 5.3).

$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$: the antisymmetric quarter-turn generator, $J^2 = -I$, the real matrix representing i throughout Chapters 9–12 and Chapter 19; identical to the real representation of i used independently by Hoffreumon and Woods [10] (Chapter 18, §18.4).

A.4 Context Graphs

$G = (V, E)$: a context graph, vertices are measurement contexts, edges carry observed bistochastic transition data (Chapter 5).

$\beta_1(G) = |E| - |V| + 1$: the cycle rank (first Betti number) of a connected graph G ; $\beta_1(G) = 0$ iff G is a tree (§5.5).

$\mathcal{R}_{\mathbb{F}}(G, d)$: the realizability space, edge-labelings of G admitting a global $\mathbb{F}$ -frame assignment (Definition 7.1).

$m_{\beta}(\mathbb{F}, \mathbb{F}'; d)$: the minimal witness, the smallest $\beta_1(G)$ over graphs G separating $\mathcal{R}_{\mathbb{F}}(G, d) \subsetneq \mathcal{R}_{\mathbb{F}'}(G, d)$ (Definition 7.2).

A.5 Recurring Abbreviations

QT: quantum theory in the ordinary complex-Hilbert-space sense (the target of the recovery layer, Part III).

RNQT: real-number quantum theory, the term used by Hoffreumon and Woods [10] for their construction (Chapter 18).

POVM: positive operator-valued measure, $\{E_i\}$ with $0 \preceq E_i \preceq I$, $\sum_i E_i = I$ (Chapter 17).

[D], [R], [A], [U]: the four-category audit tags of Chapter 18 — derived, equivalent reformulation, independently assumed, unresolved.

Appendix B

Quaternionic Preliminaries

Chapters 21 and 24 used quaternionic matrix algebra without formally collecting its conventions in one place. This appendix does so, stating precisely the facts those chapters relied on — including the two that are not true by naive analogy with the complex case, which is exactly where the earlier chapters found real difficulty.

B.1 The Quaternion Algebra

A quaternion is $q = a + bi + cj + dk$ with $a, b, c, d \in \mathbb{R}$ and $i^2 = j^2 = k^2 = ijk = -1$. Quaternion multiplication is associative but not commutative ($ij = k \neq -k = ji$). The conjugate is $\bar{q} = a - bi - cj - dk$, and the norm is $|q|^2 = q\bar{q} = a^2 + b^2 + c^2 + d^2 \in \mathbb{R}_{\geq 0}$. Every nonzero quaternion has a multiplicative inverse $q^{-1} = \bar{q}/|q|^2$ — the norm’s multiplicativity, $|pq| = |p||q|$, is what makes $\mathbb{H}$ a genuine division algebra, not merely a ring. The group of unit quaternions ($|q| = 1$) is denoted $\mathrm{Sp}(1)$; it is a three-real-dimensional Lie group, isomorphic to $SU(2)$, in contrast to the complex case’s one-dimensional $U(1)$.

Fact B.1 (Frobenius’ theorem). $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{H}$ are the only finite-dimensional associative division algebras over $\mathbb{R}$. Relaxing associativity admits exactly one further case, the octonions $\mathbb{O}$ (not used anywhere in this book).

Fact B.2 (Scalar–vector decomposition). Writing $q = (r, \mathbf{v})$ with $r \in \mathbb{R}$ and $\mathbf{v} \in \mathbb{R}^3$ (identifying $bi + cj + dk$ with (b, c, d)), quaternion multiplication becomes

$$(r_1, \mathbf{v}_1)(r_2, \mathbf{v}_2) = (r_1r_2 - \mathbf{v}_1 \cdot \mathbf{v}_2, r_1\mathbf{v}_2 + r_2\mathbf{v}_1 + \mathbf{v}_1 \times \mathbf{v}_2), \quad (\text{B.1})$$

verified directly by expanding both sides in i, j, k . The familiar dot and cross products of $\mathbb{R}^3$ are not independent constructions; they are the symmetric and antisymmetric parts of a single quaternion product.

B.2 Unit Quaternions and Rotations

Fact B.3 (The rotation formula). For a unit vector $\mathbf{n} \in \mathbb{R}^3$ and angle θ , let $q = \cos \theta + \mathbf{n} \sin \theta \in \mathrm{Sp}(1)$ (using Fact B.2’s identification). For a pure-vector quaternion $v = (0, \mathbf{v})$, $v \mapsto qvq^{-1}$ rotates $\mathbf{v}$ about axis $\mathbf{n}$ by angle 2θ .

This was checked computationally (not merely cited): for $\mathbf{n}$ the z-axis and $\theta = \pi/6$, qvq^{-1} applied to the unit vector along x reproduces the standard 60° rotation about z exactly.

Fact B.4 ($\mathrm{Sp}(1)$ double-covers $SO(3)$). The map $q \mapsto (\mathbf{v} \mapsto qvq^{-1})$ sends $\mathrm{Sp}(1) \cong SU(2) \cong S^3$ onto $SO(3)$, two-to-one: q and $-q$ act identically on every vector, since $(-q)v(-q)^{-1} = qvq^{-1}$. This is the same $\mathrm{Sp}(1)$ that appeared as the local phase group in Chapter 24, and the

same double-cover structure underlies the 720° return of spin- $\frac{1}{2}$ systems in ordinary quantum mechanics — a fact this book does not derive or otherwise depend on, recorded here only because the same group appeared for an unrelated reason (MUB counting) and the coincidence is worth naming.

B.3 The Clifford Algebra Correspondence

Fact B.5. $\mathbb{H}$ is isomorphic to the even subalgebra of the Clifford algebra of $\mathbb{R}^3$: $\mathbb{H} \cong Cl_{3,0}^+(\mathbb{R})$. Under this isomorphism the basis elements i, j, k correspond to the three coordinate bivectors e_2e_3, e_3e_1, e_1e_2 — oriented coordinate planes, not an extra spatial direction. This is standard material in geometric (Clifford) algebra, not a claim original to this book.

Remark B.6. Fact B.5 reframes what "imaginary" means for quaternions: i, j, k generate rotations in the yz, zx, xy planes respectively, and quaternion multiplication is fundamentally about composing such rotations rather than an extension of ordinary arithmetic by a mysterious extra unit. This is consistent with, and gives a coordinate-free gloss on, Chapter 19's identification of the complex unit with the generator of a plane rotation ($J = R(\pi/2)$): the quaternionic case is the same idea with three independent planes instead of one.

B.4 Quaternionic Matrices

A quaternionic $n \times n$ matrix M has entries in $\mathbb{H}$; matrix multiplication is defined exactly as over a commutative ring, and remains associative (inherited from associativity of quaternion multiplication) though the entries do not commute. The conjugate transpose is $M_{ij}^\dagger := \overline{M_{ji}}$; M is Hermitian if $M^\dagger = M$, which forces every diagonal entry to be real (a self-conjugate quaternion is real). M is unitary if $M^\dagger M = I$.

Fact B.7 (Associativity suffices for graph propagation). Every argument in Chapter 5–7 building a global frame from local transitions (tree propagation, cycle closure) uses only associativity of matrix multiplication together with the existence and well-definedness of $\pi(L)_{ij} := |L_{ij}|_{\mathbb{H}}^2 = L_{ij}\overline{L_{ij}}$. Neither requires commutativity, so these arguments transport to $\mathbb{H}$ unchanged (Chapter 21, Proposition 21.1).

B.5 Two Traps

Fact B.8 (Trace is not cyclic). For general quaternionic matrices A, B , $\text{Tr}(AB) \neq \text{Tr}(BA)$ — verified directly by example in Chapter 24. This is the genuine obstruction to a naive generalization of complex trace-based arguments, not the $\text{Sp}(1)$ -versus- $U(1)$ phase freedom originally suspected to be the difficulty (Chapter 21's closing remark).

Fact B.9 (The real part is cyclic). Despite Fact B.8, $\text{Re}[\text{Tr}(AB)] = \text{Re}[\text{Tr}(BA)]$ always. This makes $\langle A, B \rangle_{\mathbb{H}} := \text{Re}[\text{Tr}(A^\dagger B)]$ the correct symmetric real inner product on the space of quaternionic matrices, and it is this inner product — not the raw quaternionic trace — that the orthogonality argument underlying Theorem 24.3 depends on (Proposition 24.2).

Fact B.10 (Projectors are $\text{Sp}(1)$ -phase-invariant). For a unit quaternion q and a normalized $|e\rangle \in \mathbb{H}^d$, $(|e\rangle q)(|e\rangle q)^\dagger = |e\rangle\langle e|$: the rank-one projector built from a state vector does not depend on which unit quaternion represents its ray, exactly as the complex case's $U(1)$ phase drops out of $|\psi\rangle\langle\psi|$. Consequently the extra two real dimensions of $\text{Sp}(1)$ freedom (compared to $U(1)$) contribute nothing to the space spanned by one orthonormal basis's traceless projectors, which remains $(d-1)$ -dimensional exactly as in the complex case (Proposition 24.1). This was the concern flagged when Chapter 21 was first written; it resolves harmlessly, and the genuine difficulty was Fact B.8 instead.

B.6 Summary Table

	Complex ($\mathbb{C}$)	Quaternionic ($\mathbb{H}$)
Local phase group	$U(1)$, 1 real dim.	$Sp(1)$, 3 real dim.
Matrix multiplication	associative, commutative	associative, non-commutative
Trace cyclicity	$\text{Tr}(AB) = \text{Tr}(BA)$	fails; $\text{Re}[\text{Tr}(AB)] = \text{Re}[\text{Tr}(BA)]$ only
$\beta_{\mathbb{F}}(d)$	d^2	$d(2d - 1)$
MUB bound	$d + 1$	$2d + 1$ (Theorem 24.3)

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